

Christiaan's landscape

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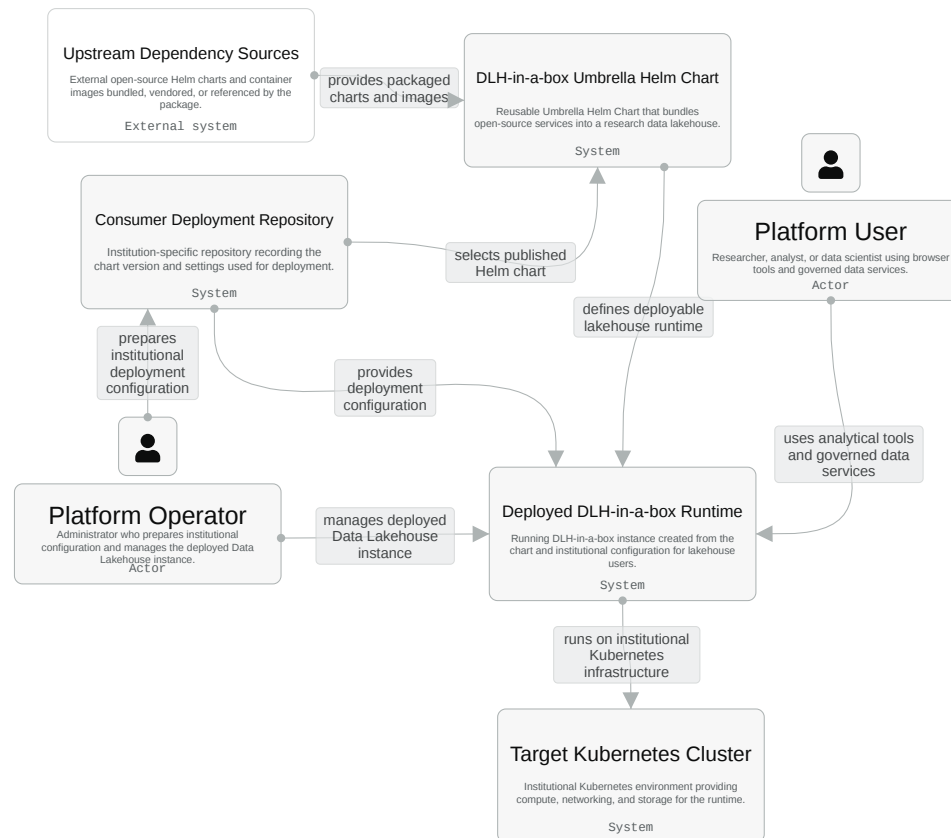
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1. Context diagrams

1.1. Context Diagram

Context diagram in the dlh-in-a-box domain.

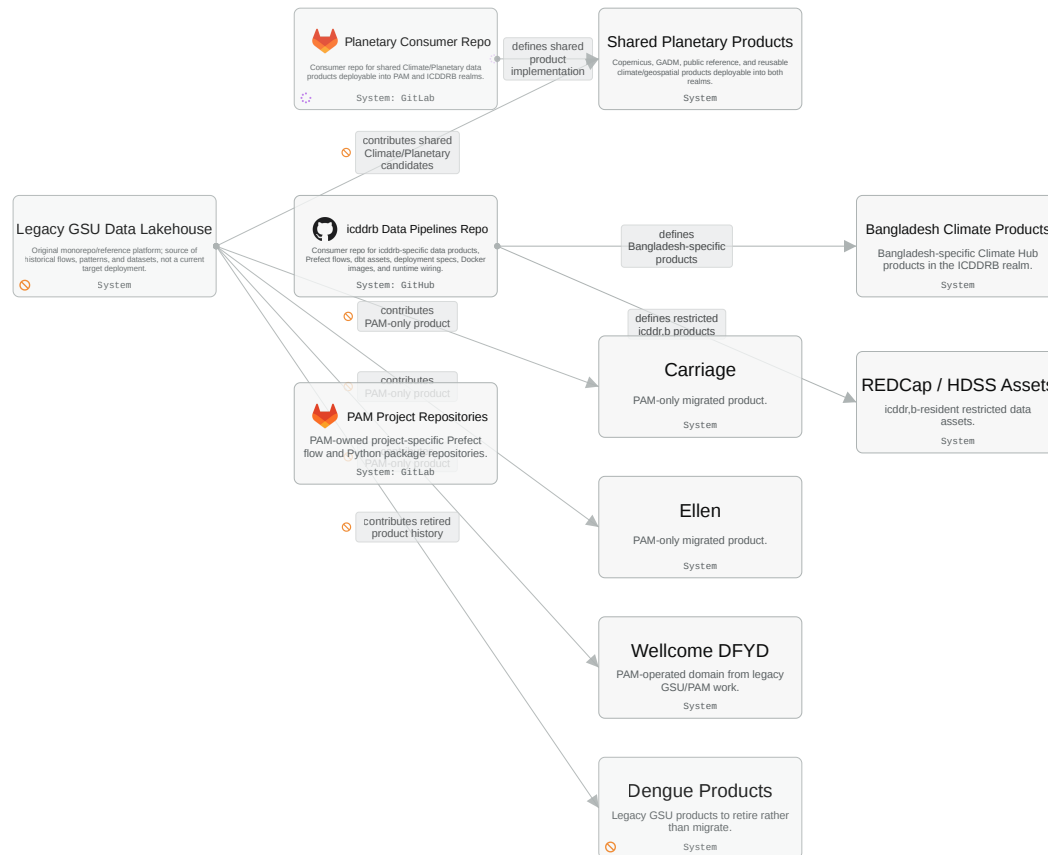
Context diagram for DLH-in-a-box. The Umbrella Helm Chart is a reusable Kubernetes package; institutional deployment repositories select a chart version and provide local settings; the resulting runtime runs on institutional Kubernetes infrastructure and exposes integrated lakehouse services to researchers.



1.2. Data Mesh Logical Overlay

Context diagram in the The PaM Data Mesh domain.

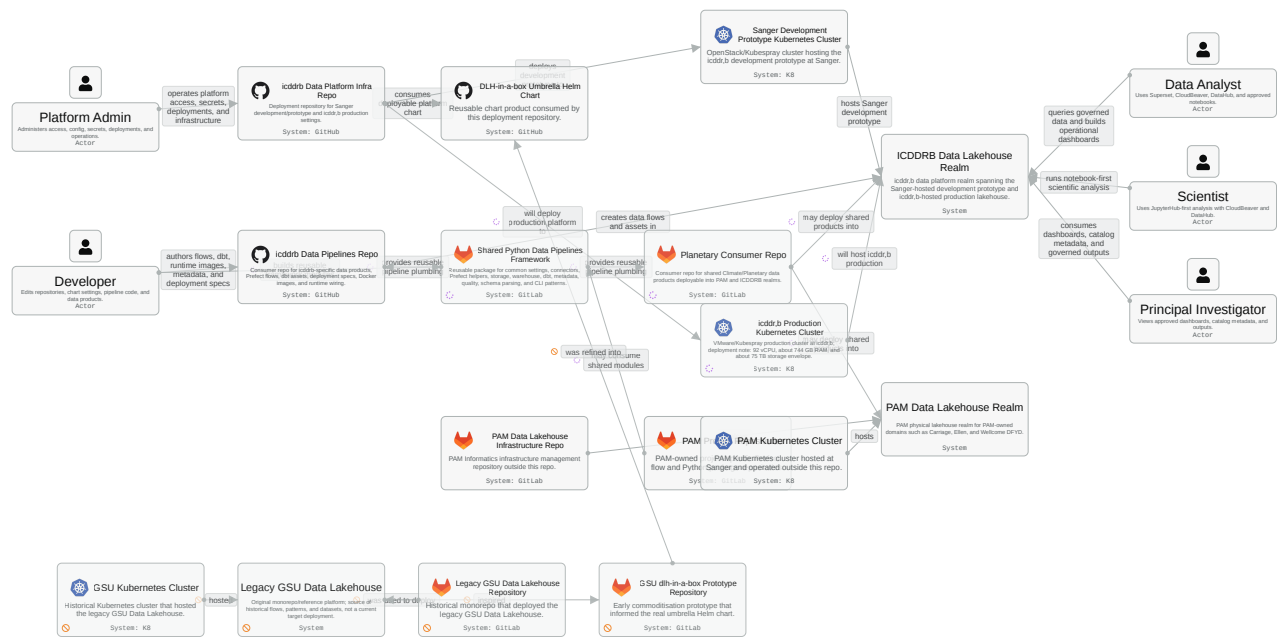
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



1.3. PaM Data Mesh Context Diagram

Context diagram in the The PaM Data Mesh domain.

Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.

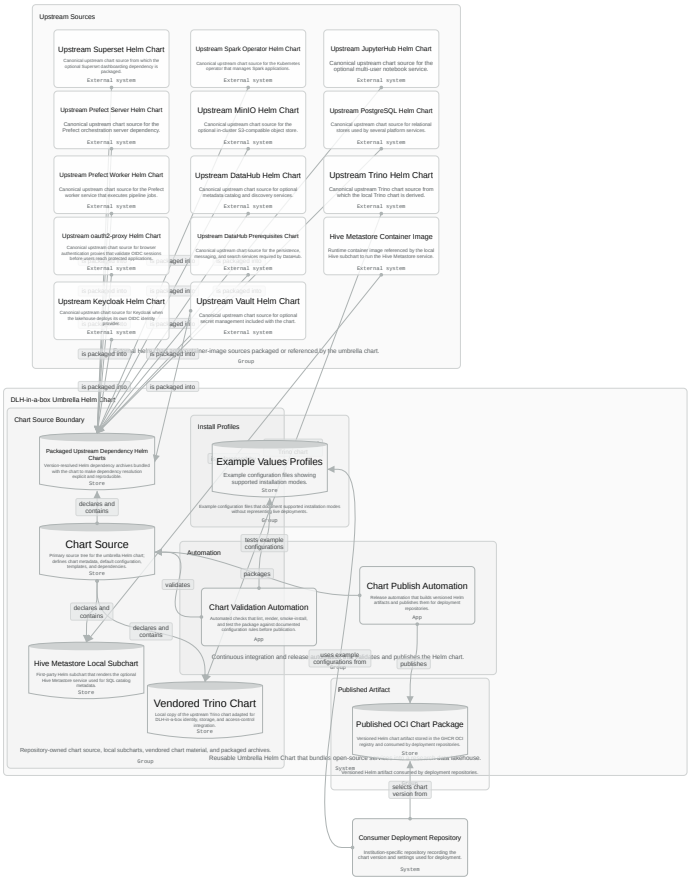


2. App diagrams

2.1. Chart Product And Packaging

DLH-in-a-box Umbrella Helm Chart app diagram in the dlh-in-a-box domain.

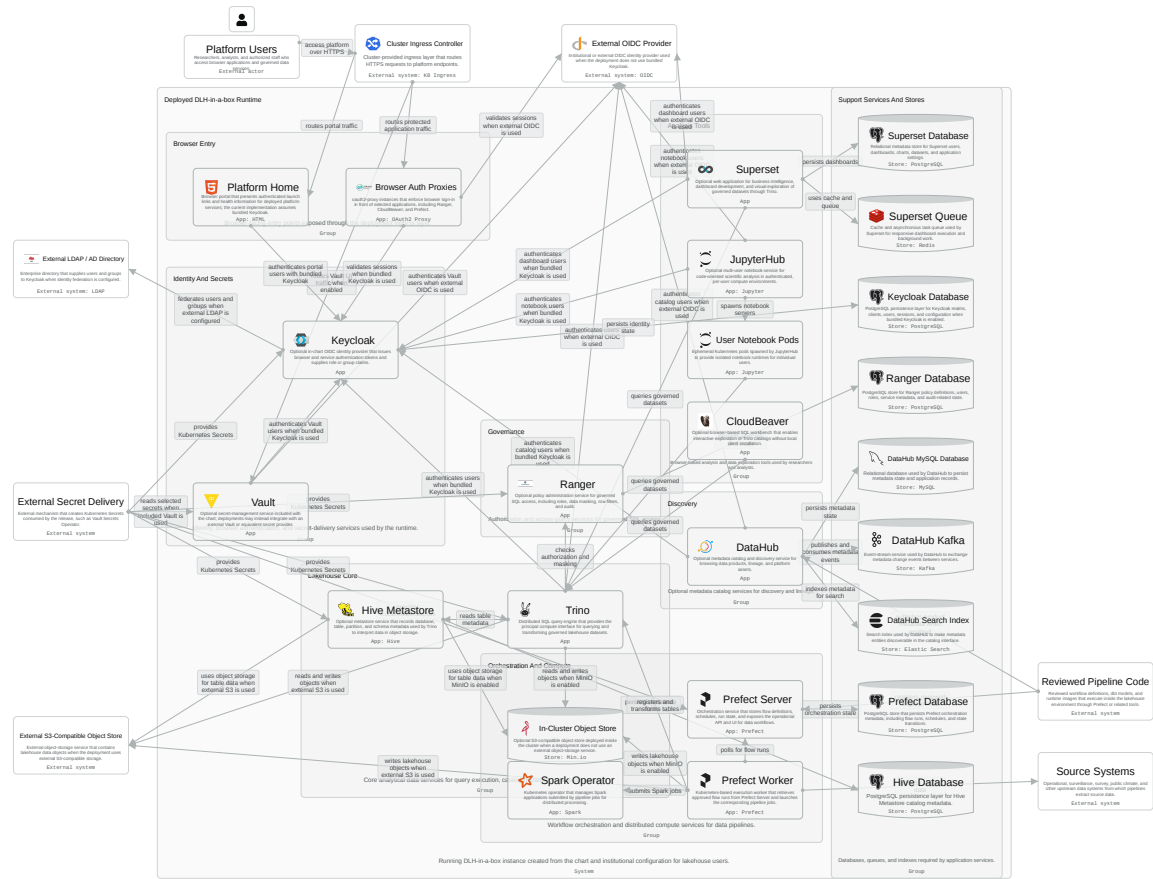
Helm chart source, packaged dependencies, examples, automation, and the published artifact used by deployment repositories.



2.2. Deployed DLH-in-a-box Runtime App Diagram

Deployed DLH-in-a-box Runtime app diagram in the dlh-in-a-box domain.

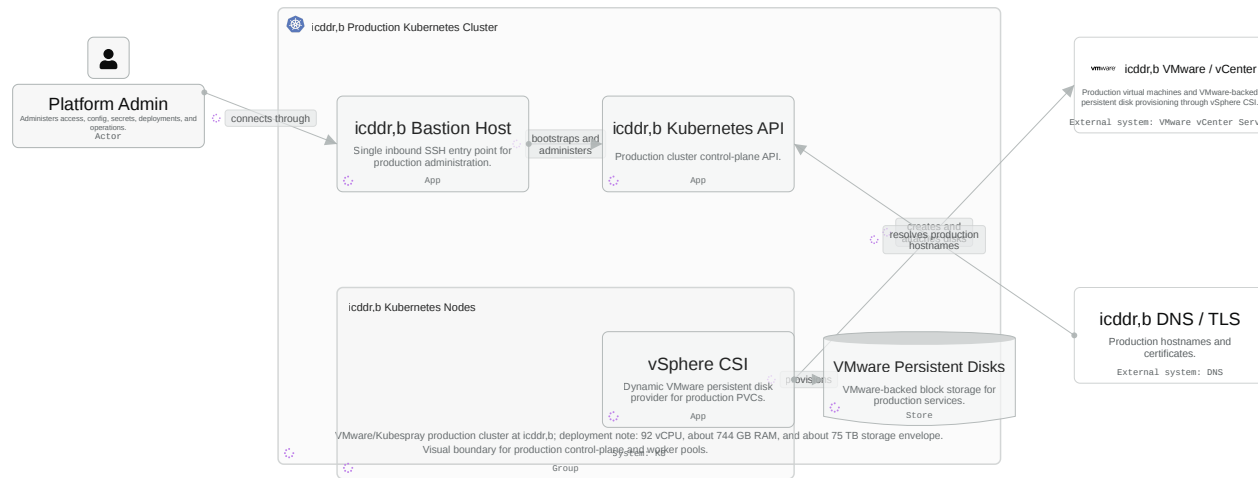
Generic DLH-in-a-box runtime deployed by the chart, including core lakehouse services, optional applications, and external context.



2.3. icddr,b Production Kubernetes Cluster App Diagram

icddr,b Production Kubernetes Cluster app diagram in the The PaM Data Mesh domain.

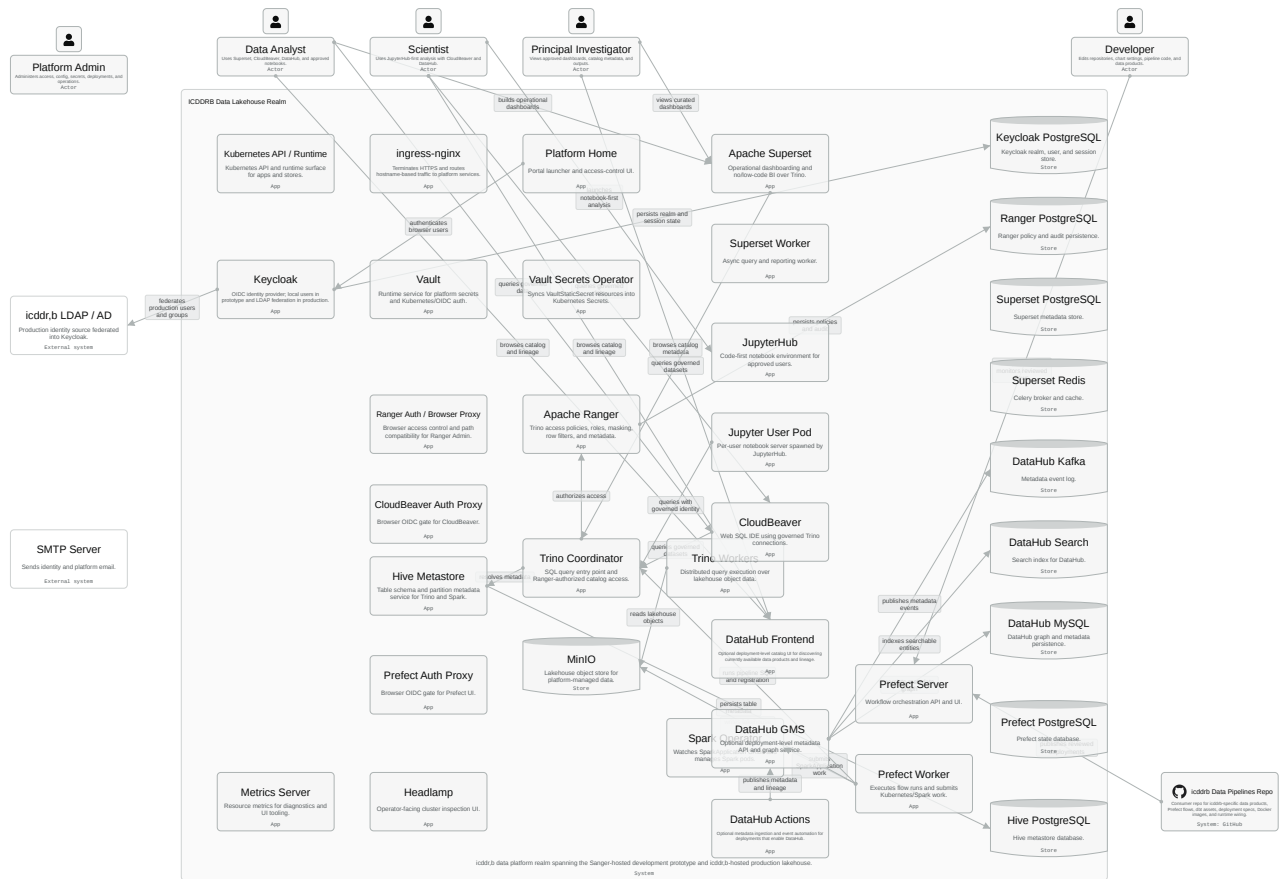
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



2.4. ICDDR Data Platform Runtime App Diagram

ICDDR Data Lakehouse Realm app diagram in the The PaM Data Mesh domain.

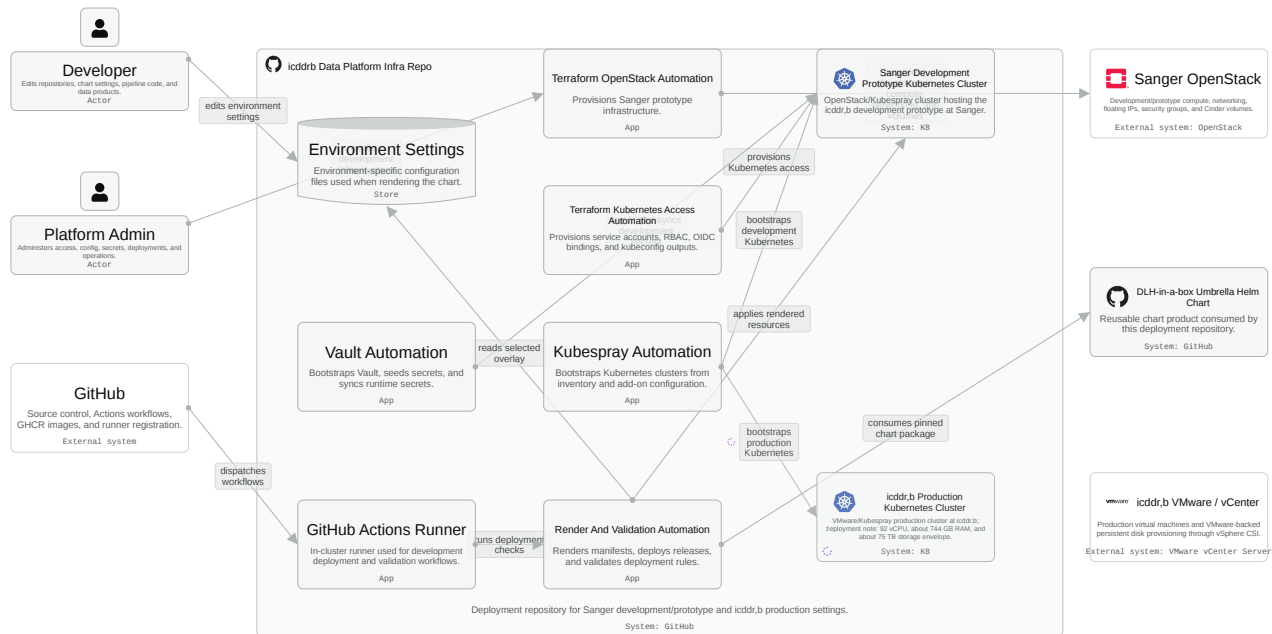
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



2.5. icddrb-data-platform-infra App Diagram

icddrb Data Platform Infra Repo app diagram in the The PaM Data Mesh domain.

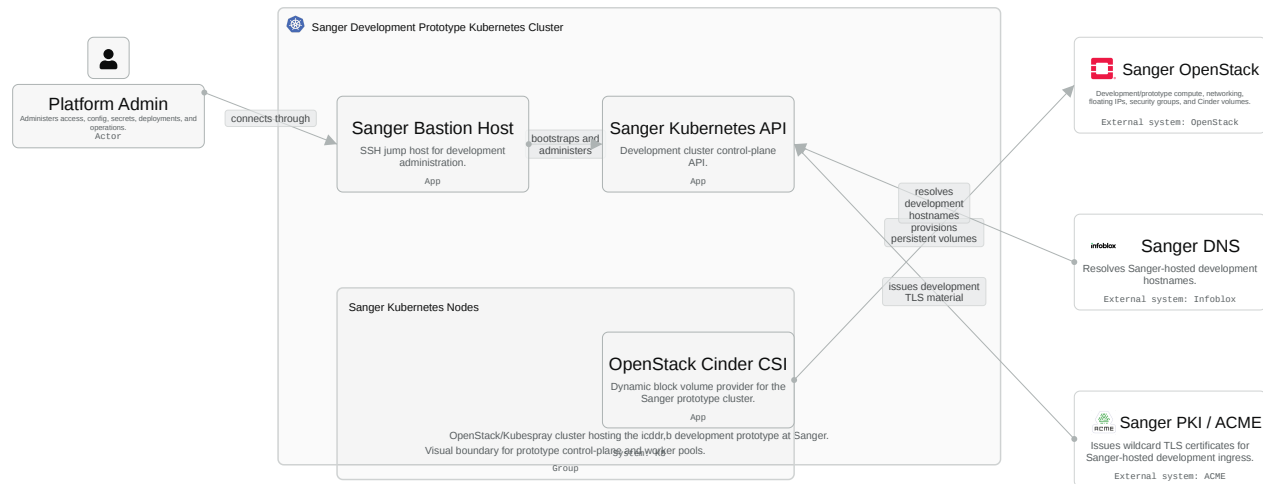
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



2.6. Sanger Development Prototype Kubernetes Cluster App Diagram

Sanger Development Prototype Kubernetes Cluster app diagram in the The PaM Data Mesh domain.

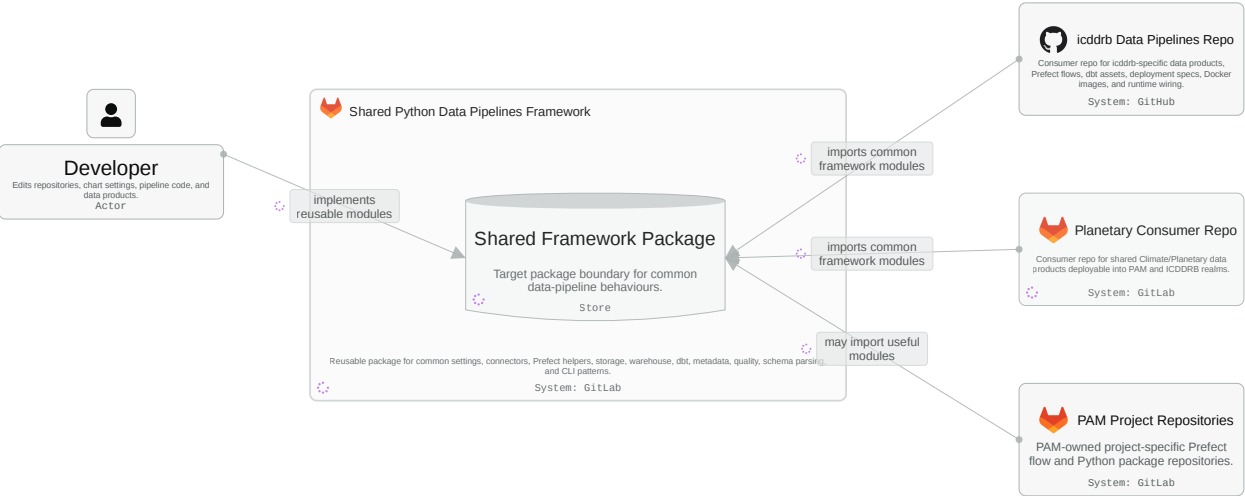
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



2.7. Shared Python Data Pipelines Framework App Diagram

Shared Python Data Pipelines Framework app diagram in the The PaM Data Mesh domain.

Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.

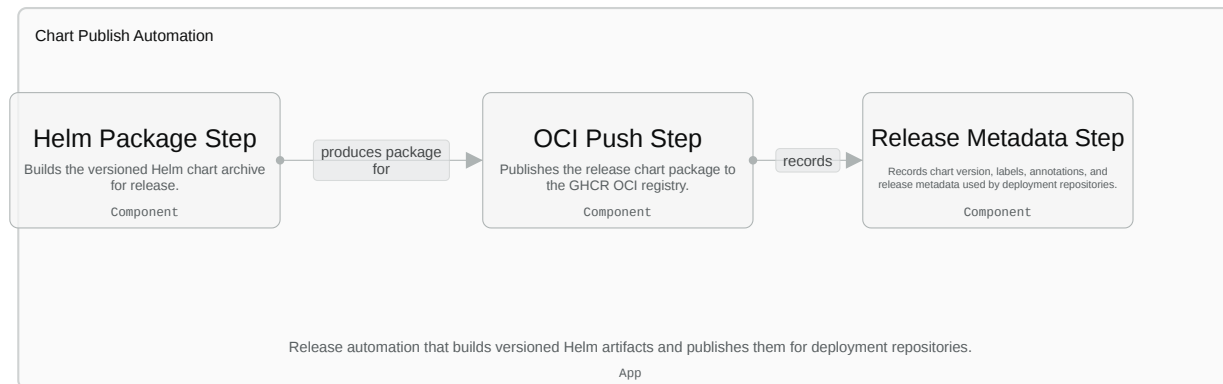


3. Component diagrams

3.1. Chart Publish Automation

Chart Publish Automation component diagram in the dlh-in-a-box domain.

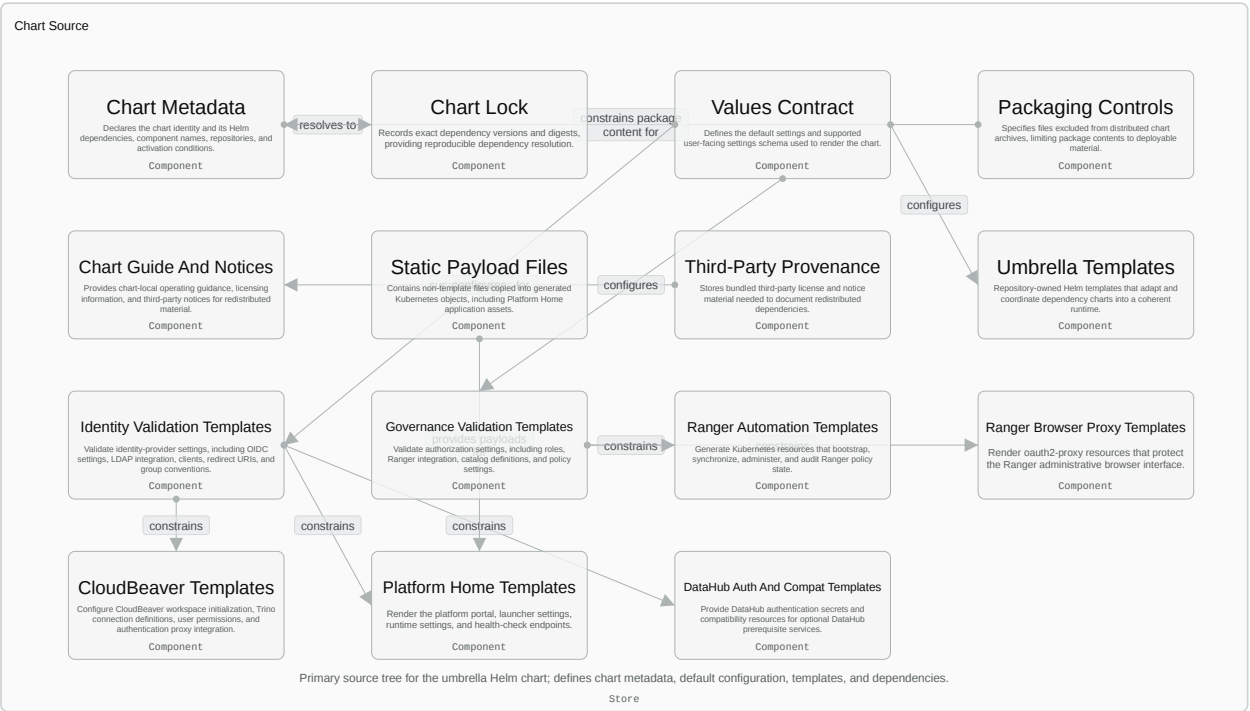
Release automation that packages and publishes the Helm artifact.



3.2. Chart Source Components

Chart Source component diagram in the dlh-in-a-box domain.

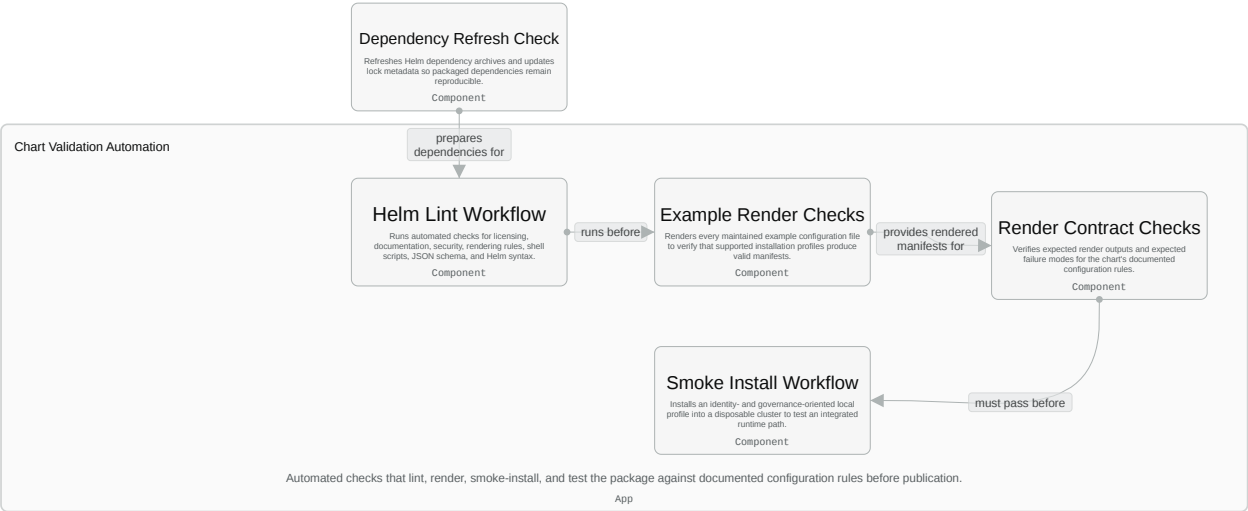
Repository-owned chart files and templates that render the deployable package.



3.3. Chart Validation Automation

Chart Validation Automation component diagram in the dlh-in-a-box domain.

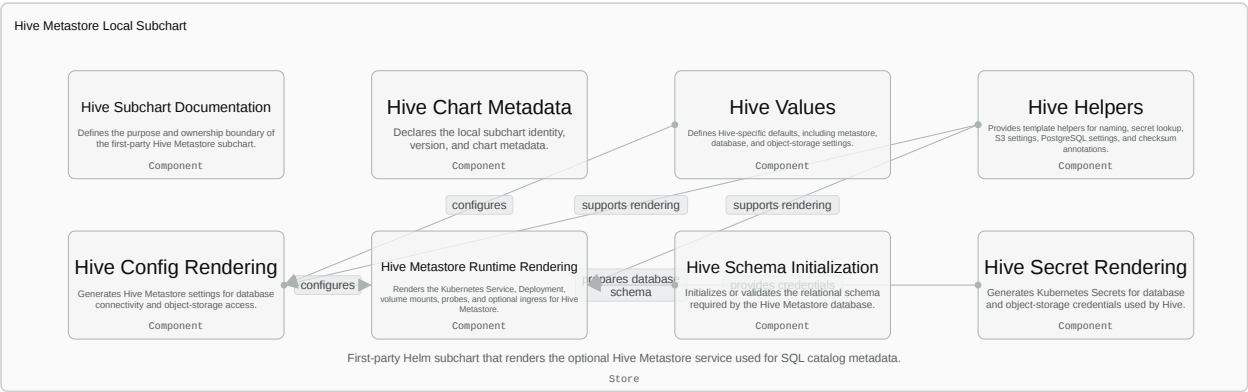
Automated checks that test the chart before publication.



3.4. Hive Metastore Local Subchart

Hive Metastore Local Subchart component diagram in the dlh-in-a-box domain.

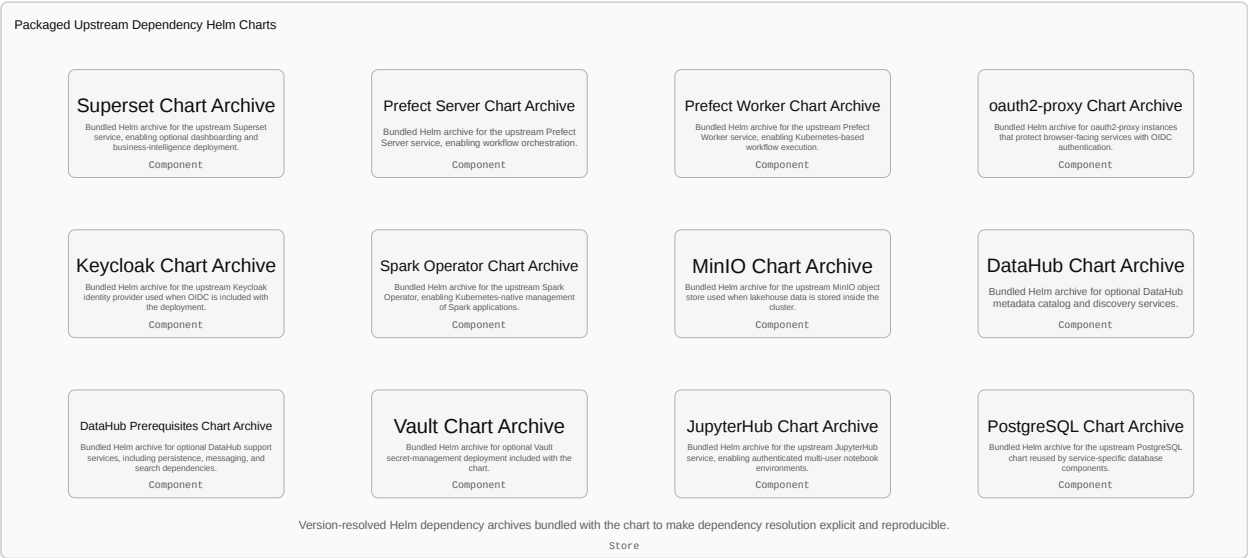
First-party Hive Metastore subchart components.



3.5. Packaged Upstream Dependencies

Packaged Upstream Dependency Helm Charts component diagram in the dlh-in-a-box domain.

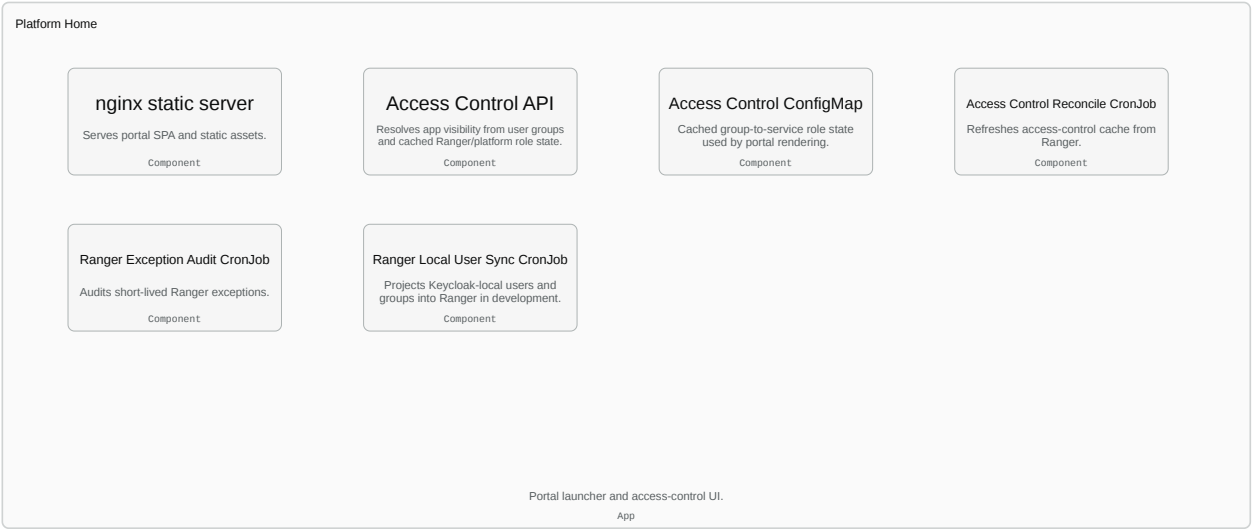
Bundled upstream Helm archives included with the umbrella chart.



3.6. Platform Home Component Diagram

Platform Home component diagram in the The PaM Data Mesh domain.

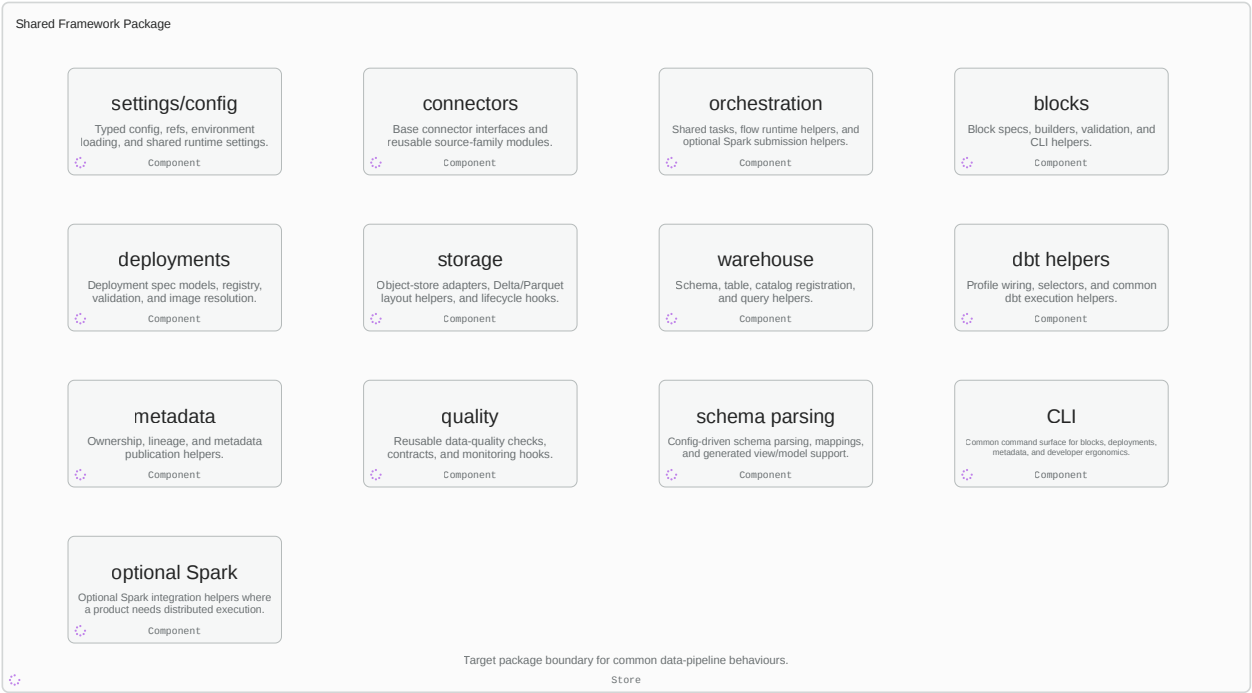
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



3.7. Shared Framework Package Component Diagram

Shared Framework Package component diagram in the The PaM Data Mesh domain.

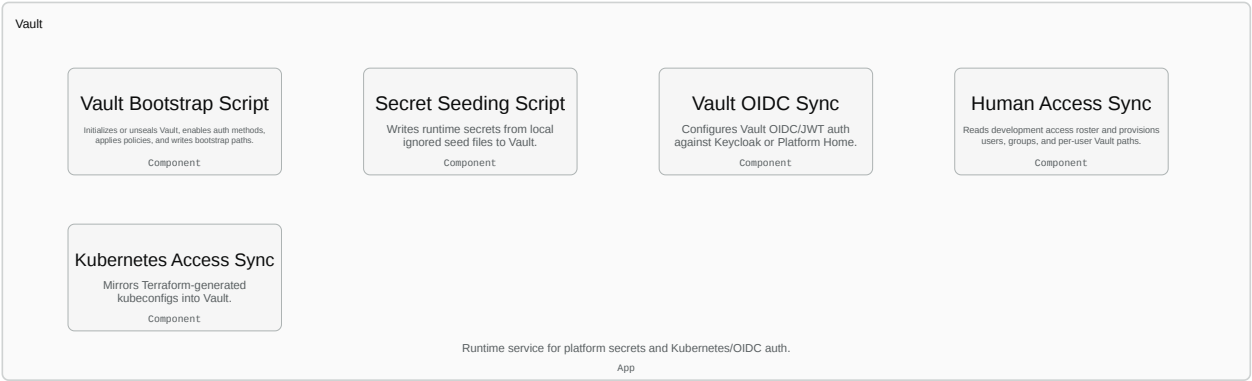
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



3.8. Vault Component Diagram

Vault component diagram in the The PaM Data Mesh domain.

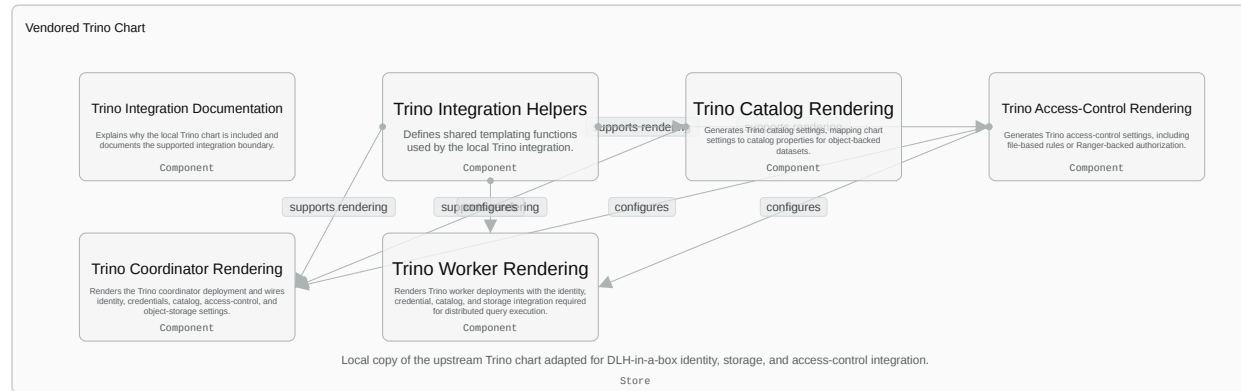
Official diagram synchronized from docs/architecture/c4-icepanel-model.xml.



3.9. Vendored Trino Chart

Vendored Trino Chart component diagram in the dlh-in-a-box domain.

Local Trino chart adaptations used for DLH-in-a-box integration.



4. Actors

4.1. Bioinformatician

Internal actor in the dlh-in-a-box domain.

Status: live

4.2. Data Analyst

Internal actor in the The PaM Data Mesh domain.

Status: live

Uses Superset, CloudBeaver, DataHub, and approved notebooks.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

Uses Superset, CloudBeaver, DataHub, and approved notebooks. Model metadata: source ID P-DATA-ANALYST; IcePanel type person or role; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

4.3. Data Engineer

Internal actor in the The PaM Data Mesh domain.

Status: live

Authors Prefect flows, dbt models, framework code, pipeline code, deployment specs, and data products; uses cluster acce

4.4. Data Engineer

Internal actor in the dlh-in-a-box domain.

Status: live

4.5. Data Scientist/ Researcher

Internal actor in the dlh-in-a-box domain.

Status: live

4.6. Developer

Internal actor in the The PaM Data Mesh domain.

Status: live

Edits repositories, chart settings, pipeline code, and data products.

Locations: Wellcome Sanger Institute, icddr,b, Cloud

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRDB Data Platform Runtime App Diagram](#)
- [icddrb-data-platform-infra App Diagram](#)
- [PaM Data Mesh Context Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

Edits repositories, chart settings, pipeline code, and data products. Model metadata: source ID P-DEVELOPER; IcePanel type person or role; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or

hosting context Wellcome Sanger Institute, icddr,b, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

4.7. Platform Admin

Internal actor in the The PaM Data Mesh domain.

Status: live

Administers access, config, secrets, deployments, and operations.

Locations: Wellcome Sanger Institute, icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)
- [ICDDRB Data Platform Runtime App Diagram](#)
- [icddrb-data-platform-infra App Diagram](#)
- [PaM Data Mesh Context Diagram](#)
- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Administers access, config, secrets, deployments, and operations. Model metadata: source ID P-ADMIN; IcePanel type person or role; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

4.8. Platform Operator

Internal actor in the dlh-in-a-box domain.

Status: live

Administrator who prepares institutional configuration and manages the deployed Data Lakehouse instance.

Locations: Wellcome Sanger Institute, icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Context Diagram](#)

Administrator who prepares institutional configuration and manages the deployed Data Lakehouse instance.

Model metadata: source ID DLH-L1-OPERATOR; IcePanel type person or role; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: Wellcome Sanger Institute, icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

4.9. Platform User

Internal actor in the dlh-in-a-box domain.

Status: live

Researcher, analyst, or data scientist using browser tools and governed data services.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [Context Diagram](#)

Researcher, analyst, or data scientist using browser tools and governed data services.

Model metadata: source ID DLH-L1-USER; IcePanel type person or role; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: icddr,b Scientists. Location or hosting context: icddr,b. Reality links: No repository

reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

4.10. Platform Users

External actor in the dlh-in-a-box domain.

Status: live

Researchers, analysts, and authorized staff who access browser applications and governed data services.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Researchers, analysts, and authorized staff who access browser applications and governed data services.

Model metadata: source ID DLH-X-USERS; IcePanel type person or role; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned.

Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: icddr,b Scientists. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

4.11. Principal Investigator

Internal actor in the The PaM Data Mesh domain.

Status: live

Views approved dashboards, catalog metadata, and outputs.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

Views approved dashboards, catalog metadata, and outputs. Model metadata: source ID P-PI; IcePanel type person or role; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

4.12. Scientist

Internal actor in the The PaM Data Mesh domain.

Status: live

Uses JupyterHub-first analysis with CloudBeaver and DataHub.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

Uses JupyterHub-first analysis with CloudBeaver and DataHub. Model metadata: source ID P-SCIENTIST; IcePanel type person or role; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5. Groups

5.1. Analysis Tools

Internal group in the dlh-in-a-box domain.

Status: live

Browser-based analysis and data exploration tools used by researchers and analysts.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Browser-based analysis and data exploration tools used by researchers and analysts.

Model metadata: source ID DLH-G-runtime-ANALYSIS-TOOLS; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.2. Automation

Internal group in the dlh-in-a-box domain.

Status: live

Continuous integration and release automation that validates and publishes the Helm chart.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: chart automation scripts](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)

Continuous integration and release automation that validates and publishes the Helm chart.

Model metadata: source ID DLH-G-chart-product-AUTOMATION; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: chart automation scripts. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.3. Browser Entry

Internal group in the dlh-in-a-box domain.

Status: live

Browser-facing entry points exposed through the deployment ingress layer.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Browser-facing entry points exposed through the deployment ingress layer.

Model metadata: source ID DLH-G-runtime-BROWSER-ENTRY; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.4. Chart Source Boundary

Internal group in the dlh-in-a-box domain.

Status: live

Repository-owned chart source, local subcharts, vendored chart material, and packaged archives.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: chart source](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)

Repository-owned chart source, local subcharts, vendored chart material, and packaged archives.

Model metadata: source ID DLH-G-chart-product-CHART-SOURCE; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: chart source. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.5. Climate / Planetary Domain

Internal group in the The PaM Data Mesh domain.

Status: live

Shared domain operated across PAM Informatics and DEI.

Locations: Wellcome Sanger Institute, icddr,b, Cloud

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger PaM Informatics
- icddr,b Scientists

Shared domain operated across PAM Informatics and DEI. Model metadata: source ID L-CLIMATE; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger PaM

Informatics, icddr,b Scientists; location or hosting context Wellcome Sanger Institute, icddr,b, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.6. Data Lakehouses

Internal group in the The PaM Data Mesh domain.

Status: live

5.7. Data Mesh

Internal group in the The PaM Data Mesh domain.

Status: live

Socio-technical operating model for distributed ownership and shared product standards.

Locations: Wellcome Sanger Institute, icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger PaM Informatics
- icddr,b Scientists

Socio-technical operating model for distributed ownership and shared product standards. Model metadata: source ID L-MESH; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger PaM Informatics, icddr,b Scientists; location or hosting context Wellcome Sanger Institute, icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.8. Data Sources

Internal group in the The PaM Data Mesh domain.

Status: live

5.9. Discovery

Internal group in the dlh-in-a-box domain.

Status: live

Optional metadata catalog services for discovery and lineage.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional metadata catalog services for discovery and lineage.

Model metadata: source ID DLH-G-runtime-DISCOVERY; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.10. Existing data pipeline deployment repositories

Internal group in the The PaM Data Mesh domain.

Status: live

5.11. Governance

Internal group in the dlh-in-a-box domain.

Status: live

Authorization and access-policy services for governed SQL access.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Authorization and access-policy services for governed SQL access.

Model metadata: source ID DLH-G-runtime-GOVERNANCE; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.12. icddr,b Core Services

Internal group in the The PaM Data Mesh domain.

Status: live

5.13. icddr,b Data Platform Users

Internal group in the The PaM Data Mesh domain.

Status: live

5.14. icddr,b Kubernetes Nodes

Internal group in the The PaM Data Mesh domain.

Status: future

Visual boundary for production control-plane and worker pools.

Locations: icddr,b

Links:

- [GitHub: production node inventory](#)

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

Visual boundary for production control-plane and worker pools. Model metadata: source ID C-PROD-NODES; IcePanel type IcePanel grouping boundary; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production node inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.15. ICDDRB Realm

Internal group in the The PaM Data Mesh domain.

Status: live

icddr,b-managed realm spanning Sanger prototype and icddr,b production.

Locations: Wellcome Sanger Institute, icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT
- icddr,b Scientists

icddr,b-managed realm spanning Sanger prototype and icddr,b production. Model metadata: source ID L-ICDDRB-REALM; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT, icddr,b Scientists; location or hosting context Wellcome Sanger Institute, icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.16. Identity And Secrets

Internal group in the dlh-in-a-box domain.

Status: live

Identity, browser authentication, and secret-delivery services used by the runtime.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Identity, browser authentication, and secret-delivery services used by the runtime.

Model metadata: source ID DLH-G-runtime-IDENTITY-AND-SECRETS; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.17. Infrastructure Repositories

Internal group in the The PaM Data Mesh domain.

Status: live

5.18. Install Profiles

Internal group in the dlh-in-a-box domain.

Status: live

Example configuration files that document supported installation modes without representing live deployments.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: example configuration files](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)

Example configuration files that document supported installation modes without representing live deployments.

Model metadata: source ID DLH-G-chart-product-INSTALL-PROFILES; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: example configuration files. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.19. Kubernetes Clusters

Internal group in the The PaM Data Mesh domain.

Status: live

5.20. Lakehouse Core

Internal group in the dlh-in-a-box domain.

Status: live

Core analytical data services for query execution, catalog metadata, and object storage.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Core analytical data services for query execution, catalog metadata, and object storage.

Model metadata: source ID DLH-G-runtime-LAKEHOUSE-CORE; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.21. Local File Dependency Charts

Internal group in the The PaM Data Mesh domain.

Status: live

5.22. OIDC-enabled browser applications

Internal group in the dlh-in-a-box domain.

Status: live

5.23. Orchestration And Compute

Internal group in the dlh-in-a-box domain.

Status: live

Workflow orchestration and distributed compute services for data pipelines.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Workflow orchestration and distributed compute services for data pipelines.

Model metadata: source ID DLH-G-runtime-ORCHESTRATION-AND-COMPUTE; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.24. Packaged Upstream Dependency Charts

Internal group in the The PaM Data Mesh domain.

Status: live

5.25. PAM Private Domains

Internal group in the The PaM Data Mesh domain.

Status: live

Carriage, Ellen, and Wellcome DFYD data products.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics

Carriage, Ellen, and Wellcome DFYD data products. Model metadata: source ID L-PAM-PRIVATE; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.26. PAM Realm

Internal group in the The PaM Data Mesh domain.

Status: live

PAM-managed physical lakehouse realm with its own environments.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics
- Sanger PaM Data Engineering and Integration

PAM-managed physical lakehouse realm with its own environments. Model metadata: source ID L-PAM-REALM; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Informatics, Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.27. Proposed new shared repositories

Internal group in the The PaM Data Mesh domain.

Status: live

5.28. Published Artifact

Internal group in the dlh-in-a-box domain.

Status: live

Versioned Helm artifact consumed by deployment repositories.

Locations: Wellcome Sanger Institute, Cloud

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)

Versioned Helm artifact consumed by deployment repositories.

Model metadata: source ID DLH-G-chart-product-PUBLISHED-ARTIFACT; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.29. REDCap / HDSS Domain

Internal group in the The PaM Data Mesh domain.

Status: live

icddr,b-resident restricted data domain.

Locations: icddr,b

Teams:

- icddr,b Scientists
- icddr,b IT

icddr,b-resident restricted data domain. Model metadata: source ID L-REDCAP-HDSS; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists, icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.30. Sanger Core Services

Internal group in the The PaM Data Mesh domain.

Status: live

Locations: Wellcome Sanger Institute

Teams:

- Sanger IDS (ISG)

5.31. Sanger Kubernetes Nodes

Internal group in the The PaM Data Mesh domain.

Status: live

Visual boundary for prototype control-plane and worker pools.

Locations: Wellcome Sanger Institute

Links:

- [GitHub: development node inventory](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Visual boundary for prototype control-plane and worker pools. Model metadata: source ID C-DEV-NODES; IcePanel type IcePanel grouping boundary; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: development node inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

5.32. Support Services and Persistent Data Stores

Internal group in the dlh-in-a-box domain.

Status: live

5.33. Support Services And Stores

Internal group in the dlh-in-a-box domain.

Status: live

Databases, queues, and indexes required by application services.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Databases, queues, and indexes required by application services.

Model metadata: source ID DLH-G-runtime-SUPPORT-SERVICES-AND-STORES; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

5.34. Upstream Sources

Internal group in the dlh-in-a-box domain.

Status: live

External Helm chart and container-image sources packaged or referenced by the umbrella chart.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: packaged dependency charts](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)

External Helm chart and container-image sources packaged or referenced by the umbrella chart.

Model metadata: source ID DLH-G-chart-product-UPSTREAM-SOURCES; IcePanel type IcePanel grouping boundary; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration.

Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: packaged dependency charts. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6. Systems

6.1. Bangladesh Climate Products

Internal system in the The PaM Data Mesh domain.

Status: live

Bangladesh-specific Climate Hub products in the ICDDRb realm.

Locations: icddr,b

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b Scientists

In groups:

- [Climate / Planetary Domain](#)
- [ICDDRb Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

Bangladesh-specific Climate Hub products in the ICDDRb realm. Model metadata: source ID L-BD-CLIMATE; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b Scientists; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.2. Bioinformatics Pipelines

Internal system in the dlh-in-a-box domain.

Status: live

6.3. Carriage

Internal system in the The PaM Data Mesh domain.

Status: live

PAM-only migrated product.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Private Domains](#)
- [PAM Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

PAM-only migrated product. Model metadata: source ID L-CARRIAGE; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.4. Cluster Ingress Controller

External system in the dlh-in-a-box domain.

Status: live

Cluster-provided ingress layer that routes HTTPS requests to platform endpoints.

Locations: icddr,b

Technologies:

- [Kubernetes Ingress](#)

Teams:

- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Cluster-provided ingress layer that routes HTTPS requests to platform endpoints.

Model metadata: source ID DLH-X-INGRESS-CONTROLLER; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.5. Consumer Deployment Repository

Internal system in the dlh-in-a-box domain.

Status: live

Institution-specific repository recording the chart version and settings used for deployment.

Locations: icddr,b, Cloud

Links:

- [GitHub: consumer deployment repository.](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Chart Product And Packaging](#)
- [Context Diagram](#)

Institution-specific repository recording the chart version and settings used for deployment.

Model metadata: source ID DLH-L1-CONSUMER-REPO; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b, Cloud. Reality links: GitHub: consumer deployment repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.6. Data Lakehouse

Internal system in the dlh-in-a-box domain.

Status: live

6.7. Dengue Products

Internal system in the The PaM Data Mesh domain.

Status: deprecated

Legacy GSU products to retire rather than migrate.

Locations: Wellcome Sanger Institute

Links:

- [GitLab: old GSU Data Lakehouse](#)

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Private Domains](#)
- [PAM Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

Legacy GSU products to retire rather than migrate. Model metadata: source ID L-DENGUE; IcePanel type data store or repository; lifecycle status Deprecated or historical object retained for context (historical object retained for context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link GitLab: old GSU Data Lakehouse. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.8. Deployed DLH-in-a-box Runtime

Internal system in the dlh-in-a-box domain.

Status: live

Running DLH-in-a-box instance created from the chart and institutional configuration for lakehouse users.

Locations: icddr,b

Links:

- [GitHub: runtime templates](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Context Diagram](#)

Running DLH-in-a-box instance created from the chart and institutional configuration for lakehouse users.

Model metadata: source ID DLH-L1-RUNTIME; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: runtime templates. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.9. DLH-in-a-box Umbrella Helm Chart

Internal system in the The PaM Data Mesh domain.

Status: live

Reusable chart product consumed by this deployment repository.

Locations: Cloud, Wellcome Sanger Institute

Technologies:

- [GitHub](#)

Links:

- [GitHub: DLH-in-a-box chart repository](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

Reusable chart product consumed by this deployment repository. Model metadata: source ID S-DLH-CHART; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-upstream); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitHub: DLH-in-a-box chart repository. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.10. DLH-in-a-box Umbrella Helm Chart

Internal system in the dlh-in-a-box domain.

Status: live

Reusable Umbrella Helm Chart that bundles open-source services into a research data lakehouse.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: dlh-in-a-box repository](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Product And Packaging](#)
- [Context Diagram](#)

Reusable Umbrella Helm Chart that bundles open-source services into a research data lakehouse.

Model metadata: source ID DLH-L1-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: dlh-in-a-box repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.11. Ellen

Internal system in the The PaM Data Mesh domain.

Status: live

PAM-only migrated product.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Private Domains](#)
- [PAM Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

PAM-only migrated product. Model metadata: source ID L-ELLEN; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.12. External LDAP / AD Directory

External system in the dlh-in-a-box domain.

Status: live

Enterprise directory that supplies users and groups to Keycloak when identity federation is configured.

Technologies:

- [LDAP](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Enterprise directory that supplies users and groups to Keycloak when identity federation is configured.

Model metadata: source ID DLH-X-LDAP; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Deployment-specific external service; the operating institution should be assigned when the concrete provider is known. Location or hosting context: Deployment-specific; this may be institutional infrastructure or a cloud-hosted service. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.13. External OIDC Provider

External system in the dlh-in-a-box domain.

Status: live

Institutional or external OIDC identity provider used when the deployment does not use bundled Keycloak.

Technologies:

- [OpenID Connect](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Institutional or external OIDC identity provider used when the deployment does not use bundled Keycloak.

Model metadata: source ID DLH-X-OIDC; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Deployment-specific external service; the operating institution should be assigned

when the concrete provider is known. Location or hosting context: Deployment-specific; this may be institutional infrastructure or a cloud-hosted service. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.14. External S3-Compatible Object Store

External system in the dlh-in-a-box domain.

Status: live

External object-storage service that contains lakehouse data objects when the deployment uses external S3-compatible storage.

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

External object-storage service that contains lakehouse data objects when the deployment uses external S3-compatible storage.

Model metadata: source ID DLH-X-OBJECT-STORAGE; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Deployment-specific external service; the operating institution should be assigned when the concrete provider is known. Location or hosting context: Deployment-specific; this may be institutional infrastructure or a cloud-hosted service. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.15. External Secret Delivery

External system in the dlh-in-a-box domain.

Status: live

External mechanism that creates Kubernetes Secrets consumed by the release, such as Vault Secrets Operator.

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

External mechanism that creates Kubernetes Secrets consumed by the release, such as Vault Secrets Operator.

Model metadata: source ID DLH-X-SECRET-SYNC; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Deployment-specific external service; the operating institution should be assigned when the concrete provider is known. Location or hosting context: Deployment-specific; this may be institutional infrastructure or a cloud-hosted service.

Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.16. GitHub

External system in the The PaM Data Mesh domain.

Status: live

Source control, Actions workflows, GHCR images, and runner registration.

Locations: Cloud

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Source control, Actions workflows, GHCR images, and runner registration. Model metadata: source ID X-GITHUB; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Externally operated service; internal ownership is limited to how the platform integrates with it; location or hosting context Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.17. GitLab

External system in the The PaM Data Mesh domain.

Status: live

Historical and PAM repository host and CI/CD source.

Locations: Cloud

Historical and PAM repository host and CI/CD source. Model metadata: source ID X-GITLAB; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Externally operated service; internal ownership is limited to how the platform integrates with it; location or hosting context Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.18. GSU dlh-in-a-box Prototype Repository

Internal system in the The PaM Data Mesh domain.

Status: deprecated

Early commoditisation prototype that informed the real umbrella Helm chart.

Locations: Wellcome Sanger Institute, Cloud

Technologies:

- [GitLab](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger PaM Informatics

In diagrams:

- [PaM Data Mesh Context Diagram](#)

Early commoditisation prototype that informed the real umbrella Helm chart. Model metadata: source ID S-GSU-DLH-IN-A-BOX-PROTOTYPE; IcePanel type system-level object; lifecycle status Deprecated or historical object retained for context (historical object retained for context); ownership or stewardship Sanger PaM Informatics, Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.19. GSU Kubernetes Cluster

Internal system in the The PaM Data Mesh domain.

Status: deprecated

Historical Kubernetes cluster that hosted the legacy GSU Data Lakehouse.

Locations: Wellcome Sanger Institute

Technologies:

- [Kubernetes](#)

Teams:

- Sanger IDS (ISG)
- Sanger PaM Informatics

In diagrams:

- [PaM Data Mesh Context Diagram](#)

Historical Kubernetes cluster that hosted the legacy GSU Data Lakehouse. Model metadata: source ID S-GSU-K8S; IcePanel type system-level object; lifecycle status Deprecated or historical object retained for context (historical object retained for context); ownership or stewardship Sanger PaM Informatics, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.20. HDSS

External system in the The PaM Data Mesh domain.

Status: live

icddr,b-resident health and demographic surveillance data source.

Locations: icddr,b

Teams:

- icddr,b IT
- icddr,b Scientists

In groups:

- [REDCap / HDSS Domain](#)

icddr,b-resident health and demographic surveillance data source. Model metadata: source ID X-HDSS; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists, icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.21. High Performance Compute Cluster

Internal system in the dlh-in-a-box domain.

Status: live

6.22. Hive Metastore Container Image

External system in the dlh-in-a-box domain.

Status: live

Runtime container image referenced by the local Hive subchart to run the Hive Metastore service.

Locations: Cloud

Links:

- [GitHub: Hive Metastore image Dockerfile](#)
- [GitHub: Hive image chart settings](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Runtime container image referenced by the local Hive subchart to run the Hive Metastore service.

Model metadata: source ID DLH-X-HIVE-IMAGE; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: Hive Metastore image Dockerfile; GitHub: Hive image chart settings. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.23. icddr,b DNS / TLS

External system in the The PaM Data Mesh domain.

Status: live

Production hostnames and certificates.

Locations: icddr,b

Technologies:

- [DNS \(Domain Name System\)](#)

Teams:

- icddr,b IT

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

Production hostnames and certificates. Model metadata: source ID X-ICDDRB-DNS; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.24. icddr,b LDAP / AD

External system in the The PaM Data Mesh domain.

Status: live

Production identity source federated into Keycloak.

Locations: icddr,b

Teams:

- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Production identity source federated into Keycloak. Model metadata: source ID X-ICDDRB-AD; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.25. icddr,b Production Data Platform

Internal system in the The PaM Data Mesh domain.

Status: future

Locations: icddr,b

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In groups:

- [Data Lakehouses](#)

6.26. icddr,b Production Kubernetes Cluster

Internal system in the The PaM Data Mesh domain.

Status: future

VMware/Kubespray production cluster at icddr,b; deployment note: 92 vCPU, about 744 GB RAM, and about 75 TB storage envelope.

Locations: icddr,b

Technologies:

- [Kubernetes](#)

Links:

- [GitHub: production cluster inventory](#)

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In groups:

- [ICDDRB Realm](#)

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)
- [icddrb-data-platform-infra App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

VMware/Kubespray production cluster at icddr,b; deployment note: 92 vCPU, about 744 GB RAM, and about 75 TB storage envelope. Model metadata: source ID S-ICDDRB-PROD-CLUSTER; IcePanel type system-level object; lifecycle status Planned or target-state object (explicitly declared); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production cluster inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.27. icddr,b Prototype Data Platform (Sanger Development Environment)

Internal system in the The PaM Data Mesh domain.

Status: live

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Data Lakehouses](#)

6.28. icddr,b TLS

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [TLS \(Transport Layer Security\)](#)

In groups:

- [icddr,b Core Services](#)

6.29. icddr,b VMware / vCenter

External system in the The PaM Data Mesh domain.

Status: live

Production virtual machines and VMware-backed persistent disk provisioning through vSphere CSI.

Locations: icddr,b

Technologies:

- [VMware vCenter Server](#)

Teams:

- icddr,b IT

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)
- [icddrb-data-platform-infra App Diagram](#)

Production virtual machines and VMware-backed persistent disk provisioning through vSphere CSI. Model metadata: source ID X-VMWARE; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.30. ICDDR Data Lakehouse Realm

Internal system in the The PaM Data Mesh domain.

Status: live

icddr,b data platform realm spanning the Sanger-hosted development prototype and icddr,b-hosted production lakehouse.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: lakehouse deployment configuration](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [ICDDRB Realm](#)

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

icddr,b data platform realm spanning the Sanger-hosted development prototype and icddr,b-hosted production lakehouse. Model metadata: source ID S-ICDDRB-LAKEHOUSE; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-target); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: lakehouse deployment configuration. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.31. icddrb Data Pipelines Repo

Internal system in the The PaM Data Mesh domain.

Status: live

Consumer repo for icddrb-specific data products, Prefect flows, dbt assets, deployment specs, Docker images, and runtime wiring.

Locations: Cloud, icddr,b

Technologies:

- [GitHub](#)

Links:

- [GitHub: icddrb-data-pipelines repository](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [ICDDRB Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)
- [ICDDRB Data Platform Runtime App Diagram](#)
- [PaM Data Mesh Context Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

Consumer repo for icddrb-specific data products, Prefect flows, dbt assets, deployment specs, Docker images, and runtime wiring. Model metadata: source ID S-ICDDRB-PIPELINES; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-downstream); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context icddr,b, Cloud; reality link GitHub: icddrb-data-pipelines repository. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.32. icddrb Data Platform Infra Repo

Internal system in the The PaM Data Mesh domain.

Status: live

Deployment repository for Sanger development/prototype and icddr,b production settings.

Locations: Wellcome Sanger Institute, icddr,b

Technologies:

- [GitHub](#)

Links:

- [GitHub: infrastructure repository](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [ICDDRB Realm](#)

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)
- [PaM Data Mesh Context Diagram](#)

Deployment repository for Sanger development/prototype and icddr,b production settings. Model metadata: source ID S-ICDDRB-PLATFORM; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: infrastructure repository. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.33. Legacy GSU Data Lakehouse

Internal system in the The PaM Data Mesh domain.

Status: deprecated

Original monorepo/reference platform; source of historical flows, patterns, and datasets, not a current target deployment.

Locations: Wellcome Sanger Institute

Links:

- [GitLab: old GSU Data Lakehouse](#)

Teams:

- Sanger PaM Informatics

In diagrams:

- [Data Mesh Logical Overlay](#)

- [PaM Data Mesh Context Diagram](#)

Original monorepo/reference platform; source of historical flows, patterns, and datasets, not a current target deployment. Model metadata: source ID S-GSU-DLH; IcePanel type system-level object; lifecycle status Deprecated or historical object retained for context (historical object retained for context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link GitLab: old GSU Data Lakehouse. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.34. Legacy GSU Data Lakehouse Repository

Internal system in the The PaM Data Mesh domain.

Status: deprecated

Historical monorepo that deployed the legacy GSU Data Lakehouse.

Locations: Wellcome Sanger Institute, Cloud

Technologies:

- [GitLab](#)

Links:

- [GitLab: old GSU Data Lakehouse repository](#)

Teams:

- Sanger PaM Informatics

In diagrams:

- [PaM Data Mesh Context Diagram](#)

Historical monorepo that deployed the legacy GSU Data Lakehouse. Model metadata: source ID S-GSU-DLH-REPO; IcePanel type system-level object; lifecycle status Deprecated or historical object retained for context (historical object retained for context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: old GSU Data Lakehouse repository. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.35. PAM Data Lakehouse Infrastructure Repo

Internal system in the The PaM Data Mesh domain.

Status: live

PAM Informatics infrastructure management repository outside this repo.

Locations: Wellcome Sanger Institute, Cloud

Technologies:

- [GitLab](#)

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Realm](#)

In diagrams:

- [PaM Data Mesh Context Diagram](#)

PAM Informatics infrastructure management repository outside this repo. Model metadata: source ID S-PAM-INFRA-REPO; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-outside-this-repo); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.36. PAM Data Lakehouse Realm

Internal system in the The PaM Data Mesh domain.

Status: live

PAM physical lakehouse realm for PAM-owned domains such as Carriage, Ellen, and Wellcome DFYD.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics
- Sanger PaM Data Engineering and Integration

In groups:

- [PAM Realm](#)

In diagrams:

- [PaM Data Mesh Context Diagram](#)

PAM physical lakehouse realm for PAM-owned domains such as Carriage, Ellen, and Wellcome DFYD. Model metadata: source ID S-PAM-LAKEHOUSE; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-outside-this-repo); ownership or stewardship Sanger PaM Informatics, Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.37. PAM Kubernetes Cluster

Internal system in the The PaM Data Mesh domain.

Status: live

PAM Kubernetes cluster hosted at Sanger and operated outside this repo.

Locations: Wellcome Sanger Institute

Technologies:

- [Kubernetes](#)

Teams:

- Sanger IDS (ISG)
- Sanger PaM Informatics

In groups:

- [PAM Realm](#)

In diagrams:

- [PaM Data Mesh Context Diagram](#)

PAM Kubernetes cluster hosted at Sanger and operated outside this repo. Model metadata: source ID S-PAM-K8S; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-outside-this-repo); ownership or stewardship Sanger PaM Informatics, Sanger IDS (ISG); location or

hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.38. PAM Project Repositories

Internal system in the The PaM Data Mesh domain.

Status: live

PAM-owned project-specific Prefect flow and Python package repositories.

Locations: Wellcome Sanger Institute, Cloud

Technologies:

- [GitLab](#)

Links:

- [GitLab: Sanger Data Lakehouse group](#)

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)
- [PaM Data Mesh Context Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

PAM-owned project-specific Prefect flow and Python package repositories. Model metadata: source ID S-PAM-PROJECT-REPOS; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current-outside-this-repo); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: Sanger Data Lakehouse group. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.39. PaM-specific Data Sources

External system in the The PaM Data Mesh domain.

Status: live

In groups:

- [Data Sources](#)

6.40. Planetary Consumer Repo

Internal system in the The PaM Data Mesh domain.

Status: future

Consumer repo for shared Climate/Planetary data products deployable into PAM and ICDDRb realms.

Locations: Wellcome Sanger Institute, icddr,b, Cloud

Technologies:

- [GitLab](#)

Teams:

- Sanger PaM Informatics
- Sanger PaM Data Engineering and Integration

In groups:

- [Climate / Planetary Domain](#)

In diagrams:

- [Data Mesh Logical Overlay](#)
- [PaM Data Mesh Context Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

Consumer repo for shared Climate/Planetary data products deployable into PAM and ICDDRb realms. Model metadata: source ID S-PLANETARY; IcePanel type system-level object; lifecycle status Planned or target-state object (normalized from target); ownership or stewardship Sanger PaM Data Engineering and Integration,

Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute, icddr,b, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.41. Project Data Capture Systems

Internal system in the dlh-in-a-box domain.

Status: live

6.42. Public Planetary Data Sources

External system in the The PaM Data Mesh domain.

Status: live

Copernicus, GADM, World Bank, HDX, NASA, GEMStat, Worldometer, and similar public data sources.

Locations: Cloud

In groups:

- [Climate / Planetary Domain](#)

Copernicus, GADM, World Bank, HDX, NASA, GEMStat, Worldometer, and similar public data sources. Model metadata: source ID X-PUBLIC-DATA; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Externally operated service; internal ownership is limited to how the platform integrates with it; location or hosting context Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.43. Rancher

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Rancher](#)

Teams:

- Sanger IDS (ISG)

In groups:

- [Sanger Core Services](#)

6.44. REDCap

External system in the The PaM Data Mesh domain.

Status: live

Source system for sample tracking and survey data.

Locations: icddr,b

Teams:

- icddr,b Scientists
- icddr,b IT

In groups:

- [REDCap / HDSS Domain](#)

Source system for sample tracking and survey data. Model metadata: source ID X-REDCAP; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists, icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.45. REDCap / HDSS Assets

Internal system in the The PaM Data Mesh domain.

Status: live

icddr,b-resident restricted data assets.

Locations: icddr,b

Links:

- [GitHub: REDCap and HDSS dbt assets](#)

Teams:

- icddr,b Scientists
- icddr,b IT

In groups:

- [ICDDR B Realm](#)
- [REDCap / HDSS Domain](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

icddr,b-resident restricted data assets. Model metadata: source ID L-REDCAP-HDSS-ASSETS; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b Scientists, icddr,b IT; location or hosting context icddr,b; reality link GitHub: REDCap and HDSS dbt assets. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.46. RedCap Production Instance

External system in the The PaM Data Mesh domain.

Status: future

Locations: icddr,b

Teams:

- icddr,b IT

In groups:

- [Data Sources](#)

6.47. RedCap Prototype Instance

External system in the The PaM Data Mesh domain.

Status: live

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics

In groups:

- [Data Sources](#)

6.48. Reviewed Pipeline Code

External system in the dlh-in-a-box domain.

Status: live

Reviewed workflow definitions, dbt models, and runtime images that execute inside the lakehouse environment through Prefect or related tools.

Locations: icddr,b, Cloud

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Reviewed workflow definitions, dbt models, and runtime images that execute inside the lakehouse environment through Prefect or related tools.

Model metadata: source ID DLH-X-PIPELINE-CODE; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b, Cloud. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.49. Sanger Ceph S3

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Ceph](#)

Teams:

- Sanger IDS (ISG)

In groups:

- [Sanger Core Services](#)

6.50. Sanger Development Prototype Kubernetes Cluster

Internal system in the The PaM Data Mesh domain.

Status: live

OpenStack/Kubespray cluster hosting the icddr,b development prototype at Sanger.

Locations: Wellcome Sanger Institute

Technologies:

- [Kubernetes](#)

Links:

- [GitHub: development cluster inventory](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In groups:

- [ICDDR B Realm](#)

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

- [PaM Data Mesh Context Diagram](#)
- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

OpenStack/Kubespray cluster hosting the icddr,b development prototype at Sanger. Model metadata: source ID S-SANGER-DEV-CLUSTER; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (normalized from current); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: development cluster inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.51. Sanger DNS

External system in the The PaM Data Mesh domain.

Status: live

Resolves Sanger-hosted development hostnames.

Locations: Wellcome Sanger Institute

Technologies:

- [Infoblox](#)

Teams:

- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Resolves Sanger-hosted development hostnames. Model metadata: source ID X-SANGER-DNS; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.52. Sanger OpenStack

External system in the The PaM Data Mesh domain.

Status: live

Development/prototype compute, networking, floating IPs, security groups, and Cinder volumes.

Locations: Wellcome Sanger Institute

Technologies:

- [OpenStack](#)

Teams:

- Sanger IDS (ISG)
- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)
- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Development/prototype compute, networking, floating IPs, security groups, and Cinder volumes. Model metadata: source ID X-OPENSTACK; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger IDS (ISG), Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.53. Sanger PKI / ACME

External system in the The PaM Data Mesh domain.

Status: live

Issues wildcard TLS certificates for Sanger-hosted development ingress.

Locations: Wellcome Sanger Institute

Technologies:

- [ACME](#)

Teams:

- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Issues wildcard TLS certificates for Sanger-hosted development ingress. Model metadata: source ID X-SANGER-PKI; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.54. Shared Planetary Products

Internal system in the The PaM Data Mesh domain.

Status: live

Copernicus, GADM, public reference, and reusable climate/geospatial products deployable into both realms.

Locations: Wellcome Sanger Institute, icddr,b, Cloud

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger PaM Informatics
- icddr,b Scientists

In groups:

- [Climate / Planetary Domain](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

Copernicus, GADM, public reference, and reusable climate/geospatial products deployable into both realms. Model metadata: source ID L-PLANETARY-SHARED; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger PaM Informatics, icddr,b Scientists; location or hosting context Wellcome Sanger Institute, icddr,b, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.55. Shared Python Data Pipelines Framework

Internal system in the The PaM Data Mesh domain.

Status: future

Reusable package for common settings, connectors, Prefect helpers, storage, warehouse, dbt, metadata, quality, schema parsing, and CLI patterns.

Locations: Wellcome Sanger Institute, Cloud

Technologies:

- [GitLab](#)

Links:

- [GitLab: Sanger Data Lakehouse packages](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [PaM Data Mesh Context Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

Reusable package for common settings, connectors, Prefect helpers, storage, warehouse, dbt, metadata, quality, schema parsing, and CLI patterns. Model metadata: source ID S-SHARED-FRAMEWORK; IcePanel type system-level object; lifecycle status Planned or target-state object (normalized from target); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: Sanger Data Lakehouse packages. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.56. SMTP Server

External system in the The PaM Data Mesh domain.

Status: live

Sends identity and platform email.

Locations: icddr,b

Teams:

- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Sends identity and platform email. Model metadata: source ID X-SMTP; IcePanel type system-level object; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship icddr,b IT; location or hosting context icddr,b; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

6.57. Source Systems

External system in the dlh-in-a-box domain.

Status: live

Operational, surveillance, survey, public climate, and other upstream data systems from which pipelines extract source data.

Locations: icddr,b

Teams:

- icddr,b Scientists

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Operational, surveillance, survey, public climate, and other upstream data systems from which pipelines extract source data.

Model metadata: source ID DLH-X-SOURCE-SYSTEMS; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: icddr,b Scientists. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.58. Target Kubernetes Cluster

Internal system in the dlh-in-a-box domain.

Status: live

Institutional Kubernetes environment providing compute, networking, and storage for the runtime.

Locations: icddr,b

Teams:

- icddr,b IT

In diagrams:

- [Context Diagram](#)

Institutional Kubernetes environment providing compute, networking, and storage for the runtime.

Model metadata: source ID DLH-L1-TARGET-CLUSTER; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: icddr,b IT. Location or hosting context: icddr,b. Reality links: No repository reality link has been assigned; this object is conceptual, deployment-specific, or external to the repository. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.59. Third Party Planetary Data Sources

Internal system in the dlh-in-a-box domain.

Status: live

6.60. Upstream DataHub Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for optional metadata catalog and discovery services.

Locations: Cloud

Links:

- [GitHub: packaged DataHub chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for optional metadata catalog and discovery services.

Model metadata: source ID DLH-X-DATAHUB-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged DataHub chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.61. Upstream DataHub Prerequisites Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the persistence, messaging, and search services required by DataHub.

Locations: Cloud

Links:

- [GitHub: packaged DataHub prerequisites chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the persistence, messaging, and search services required by DataHub.

Model metadata: source ID DLH-X-DATAHUB-PREREQS-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged DataHub prerequisites chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.62. Upstream Dependency Sources

External system in the dlh-in-a-box domain.

Status: live

External open-source Helm charts and container images bundled, vendored, or referenced by the package.

Locations: Cloud

Links:

- [GitHub: packaged dependency charts](#)

In diagrams:

- [Context Diagram](#)

External open-source Helm charts and container images bundled, vendored, or referenced by the package.

Model metadata: source ID DLH-L1-UPSTREAM-DEPS; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No single internal owner is assigned because the object represents multiple externally maintained open-source projects. Location or hosting context: Cloud. Reality links: GitHub: packaged dependency charts. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.63. Upstream Hive Metastore Docker Container

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Docker](#)

6.64. Upstream JupyterHub Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the optional multi-user notebook service.

Locations: Cloud

Links:

- [GitHub: packaged JupyterHub chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the optional multi-user notebook service.

Model metadata: source ID DLH-X-JUPYTERHUB-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged JupyterHub chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.65. Upstream Keycloak Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for Keycloak when the lakehouse deploys its own OIDC identity provider.

Locations: Cloud

Links:

- [GitHub: packaged Keycloak chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for Keycloak when the lakehouse deploys its own OIDC identity provider.

Model metadata: source ID DLH-X-KEYCLOAK-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream

project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Keycloak chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.66. Upstream MinIO Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the optional in-cluster S3-compatible object store.

Locations: Cloud

Links:

- [GitHub: packaged MinIO chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the optional in-cluster S3-compatible object store.

Model metadata: source ID DLH-X-MINIO-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged MinIO chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.67. Upstream oauth2-proxy Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for browser authentication proxies that validate OIDC sessions before users reach protected applications.

Locations: Cloud

Links:

- [GitHub: packaged oauth2-proxy chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for browser authentication proxies that validate OIDC sessions before users reach protected applications.

Model metadata: source ID DLH-X-OAUTH2-PROXY-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged oauth2-proxy chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.68. Upstream PostgreSQL Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for relational stores used by several platform services.

Locations: Cloud

Links:

- [GitHub: packaged PostgreSQL chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for relational stores used by several platform services.

Model metadata: source ID DLH-X-POSTGRES-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the

upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged PostgreSQL chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.69. Upstream Prefect Server Chart

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Helm](#)

6.70. Upstream Prefect Server Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the Prefect orchestration server dependency.

Locations: Cloud

Links:

- [GitHub: packaged Prefect Server chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the Prefect orchestration server dependency.

Model metadata: source ID DLH-X-PREFECT-SERVER-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Prefect Server chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.71. Upstream Prefect Worker Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the Prefect worker service that executes pipeline jobs.

Locations: Cloud

Links:

- [GitHub: packaged Prefect Worker chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the Prefect worker service that executes pipeline jobs.

Model metadata: source ID DLH-X-PREFECT-WORKER-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Prefect Worker chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.72. Upstream Spark Operator Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for the Kubernetes operator that manages Spark applications.

Locations: Cloud

Links:

- [GitHub: packaged Spark Operator chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for the Kubernetes operator that manages Spark applications.

Model metadata: source ID DLH-X-SPARK-OPERATOR-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Spark Operator chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.73. Upstream Superset Chart

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Helm](#)

6.74. Upstream Superset Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source from which the optional Superset dashboarding dependency is packaged.

Locations: Cloud

Links:

- [GitHub: packaged Superset chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source from which the optional Superset dashboarding dependency is packaged.

Model metadata: source ID DLH-X-SUPERSET-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Superset chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.75. Upstream Trino Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream Trino chart source from which the local Trino chart is derived.

Locations: Cloud

Links:

- [GitHub: vendored Trino chart source](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream Trino chart source from which the local Trino chart is derived.

Model metadata: source ID DLH-X-TRINO-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: vendored Trino chart source. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.76. Upstream Trino Helm Chart

External system in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Helm](#)

6.77. Upstream Vault Helm Chart

External system in the dlh-in-a-box domain.

Status: live

Canonical upstream chart source for optional secret management included with the chart.

Locations: Cloud

Links:

- [GitHub: packaged Vault chart](#)

In groups:

- [Upstream Sources](#)

In diagrams:

- [Chart Product And Packaging](#)

Canonical upstream chart source for optional secret management included with the chart.

Model metadata: source ID DLH-X-VAULT-CHART; IcePanel type system-level object; Top-level object in the DLH-in-a-box domain. IcePanel group membership: Upstream Sources. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: No internal IcePanel owner is assigned because the upstream project is externally maintained; Sanger PaM Data Engineering and Integration stewards the packaging decision. Location or hosting context: Cloud. Reality links: GitHub: packaged Vault chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

6.78. Wellcome DFYD

Internal system in the The PaM Data Mesh domain.

Status: live

PAM-operated domain from legacy GSU/PAM work.

Locations: Wellcome Sanger Institute

Teams:

- Sanger PaM Informatics

In groups:

- [PAM Private Domains](#)
- [PAM Realm](#)

In diagrams:

- [Data Mesh Logical Overlay](#)

PAM-operated domain from legacy GSU/PAM work. Model metadata: source ID L-DFYD; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred live from the current model context); ownership or stewardship Sanger PaM Informatics; location or hosting context Wellcome Sanger Institute; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7. Apps

7.1. Apache Ranger

Internal app in the The PaM Data Mesh domain.

Status: live

Trino access policies, roles, masking, row filters, and metadata.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Ranger admin template](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Trino access policies, roles, masking, row filters, and metadata. Model metadata: source ID C-RANGER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Ranger admin template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.2. Apache Superset

Internal app in the The PaM Data Mesh domain.

Status: live

Operational dashboarding and no/low-code BI over Trino.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Superset chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Operational dashboarding and no/low-code BI over Trino. Model metadata: source ID C-SUPERSET; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Superset chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.3. Browser Auth Proxies

Internal app in the dlh-in-a-box domain.

Status: live

oauth2-proxy instances that enforce browser sign-in in front of selected applications, including Ranger, CloudBeaver, and Prefect.

Locations: icddr,b

Technologies:

- [OAuth2 Proxy](#)

Links:

- [GitHub: oauth2-proxy chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Browser Entry](#)
- [OIDC-enabled browser applications](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

oauth2-proxy instances that enforce browser sign-in in front of selected applications, including Ranger, CloudBeaver, and Prefect.

Model metadata: source ID DLH-R-AUTH-PROXIES; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME).

IcePanel group membership: Browser Entry. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: oauth2-proxy chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.4. Chart Publish Automation

Internal app in the dlh-in-a-box domain.

Status: live

Release automation that builds versioned Helm artifacts and publishes them for deployment repositories.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/workflows/helm-publish.yaml](#)
- [GitHub: hack/package.sh](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Automation](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Chart Publish Automation](#)

Release automation that builds versioned Helm artifacts and publishes them for deployment repositories.

Model metadata: source ID DLH-C2-PUBLISH; IcePanel type application or executable service; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Automation. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/workflows/helm-publish.yaml; GitHub: hack/package.sh. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.5. Chart Validation Automation

Internal app in the dlh-in-a-box domain.

Status: live

Automated checks that lint, render, smoke-install, and test the package against documented configuration rules before publication.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/workflows/helm-lint.yaml](#)
- [GitHub: .github/workflows/helm-smoke-install.yaml](#)
- [GitHub: hack](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Automation](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Chart Validation Automation](#)

Automated checks that lint, render, smoke-install, and test the package against documented configuration rules before publication.

Model metadata: source ID DLH-C2-VALIDATION; IcePanel type application or executable service; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART).

IcePanel group membership: Automation. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/workflows/helm-lint.yaml; GitHub: .github/workflows/helm-smoke-install.yaml; GitHub: hack. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.6. CloudBeaver

Internal app in the The PaM Data Mesh domain.

Status: live

Web SQL IDE using governed Trino connections.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: CloudBeaver template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Web SQL IDE using governed Trino connections. Model metadata: source ID C-CLOUDBEAVER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: CloudBeaver template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.7. CloudBeaver

Internal app in the dlh-in-a-box domain.

Status: live

Optional browser-based SQL workbench that enables interactive exploration of Trino catalogs without local client installation.

Locations: icddr,b

Technologies:

- [CloudBeaver](#)

Links:

- [GitHub: CloudBeaver template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Analysis Tools](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional browser-based SQL workbench that enables interactive exploration of Trino catalogs without local client installation.

Model metadata: source ID DLH-R-CLOUDBEAVER; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Analysis Tools. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: CloudBeaver template. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.8. CloudBeaver Auth Proxy

Internal app in the The PaM Data Mesh domain.

Status: live

Browser OIDC gate for CloudBeaver.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: oauth2-proxy chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Browser OIDC gate for CloudBeaver. Model metadata: source ID C-CLOUDBEAVER-PROXY; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: oauth2-proxy chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.9. DataHub

Internal app in the dlh-in-a-box domain.

Status: live

Optional metadata catalog and discovery service for browsing data products, lineage, and platform assets.

Locations: icddr,b

Technologies:

- [DataHub](#)

Links:

- [GitHub: DataHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Discovery](#)
- [OIDC-enabled browser applications](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional metadata catalog and discovery service for browsing data products, lineage, and platform assets.

Model metadata: source ID DLH-R-DATAHUB; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Discovery. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: DataHub chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.10. DataHub Actions

Internal app in the The PaM Data Mesh domain.

Status: live

Optional metadata ingestion and event automation for deployments that enable DataHub.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Optional metadata ingestion and event automation for deployments that enable DataHub. Model metadata: source ID C-DATAHUB-ACTIONS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.11. DataHub Frontend

Internal app in the The PaM Data Mesh domain.

Status: live

Optional deployment-level catalog UI for discovering currently available data products and lineage.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Optional deployment-level catalog UI for discovering currently available data products and lineage. Model metadata: source ID C-DATAHUB-FE; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.12. DataHub GMS

Internal app in the The PaM Data Mesh domain.

Status: live

Optional deployment-level metadata API and graph service.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Optional deployment-level metadata API and graph service. Model metadata: source ID C-DATAHUB-GMS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.13. GitHub Actions Runner

Internal app in the The PaM Data Mesh domain.

Status: live

In-cluster runner used for development deployment and validation workflows.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: k8s/kubespray/addons/github-runner](https://github.com/k8s/kubespray/addons/github-runner)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

In-cluster runner used for development deployment and validation workflows. Model metadata: source ID C-GHA-RUNNER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: k8s/kubespray/addons/github-runner. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.14. Headlamp

Internal app in the The PaM Data Mesh domain.

Status: live

Operator-facing cluster inspection UI.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Headlamp addon](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Operator-facing cluster inspection UI. Model metadata: source ID C-HEADLAMP; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Headlamp addon. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.15. Hive Metastore

Internal app in the dlh-in-a-box domain.

Status: live

Optional metastore service that records database, table, partition, and schema metadata used by Trino to interpret data in object storage.

Locations: icddr,b

Technologies:

- [Apache Hive](#)

Links:

- [GitHub: Hive Metastore subchart](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Lakehouse Core](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional metastore service that records database, table, partition, and schema metadata used by Trino to interpret data in object storage.

Model metadata: source ID DLH-R-HIVE; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Lakehouse Core. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Hive Metastore subchart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.16. Hive Metastore

Internal app in the The PaM Data Mesh domain.

Status: live

Table schema and partition metadata service for Trino and Spark.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Hive Metastore subchart](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Table schema and partition metadata service for Trino and Spark. Model metadata: source ID C-HIVE; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Hive Metastore subchart. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.17. icddr,b Bastion Host

Internal app in the The PaM Data Mesh domain.

Status: future

Single inbound SSH entry point for production administration.

Locations: icddr,b

Links:

- [GitHub: production inventory](#)

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

Single inbound SSH entry point for production administration. Model metadata: source ID C-PROD-BASTION; IcePanel type application or executable service; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.18. icddr,b Kubernetes API

Internal app in the The PaM Data Mesh domain.

Status: future

Production cluster control-plane API.

Locations: icddr,b

Links:

- [GitHub: production cluster inventory](#)

Teams:

- icddr,b IT

- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

Production cluster control-plane API. Model metadata: source ID C-PROD-K8S-API; IcePanel type application or executable service; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production cluster inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.19. ingress-nginx

Internal app in the The PaM Data Mesh domain.

Status: live

Terminates HTTPS and routes hostname-based traffic to platform services.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: ingress-nginx addon](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Terminates HTTPS and routes hostname-based traffic to platform services. Model metadata: source ID C-INGRESS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: ingress-nginx addon. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.20. Jupyter User Pod

Internal app in the The PaM Data Mesh domain.

Status: live

Per-user notebook server spawned by JupyterHub.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: JupyterHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Per-user notebook server spawned by JupyterHub. Model metadata: source ID C-JUPYTER-POD; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: JupyterHub chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.21. JupyterHub

Internal app in the dlh-in-a-box domain.

Status: live

Optional multi-user notebook service for code-oriented scientific analysis in authenticated, per-user compute environments.

Locations: icddr,b

Technologies:

- [Jupyter](#)

Links:

- [GitHub: JupyterHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Analysis Tools](#)
- [OIDC-enabled browser applications](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional multi-user notebook service for code-oriented scientific analysis in authenticated, per-user compute environments.

Model metadata: source ID DLH-R-JUPYTERHUB; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Analysis Tools. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: JupyterHub chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.22. JupyterHub

Internal app in the The PaM Data Mesh domain.

Status: live

Code-first notebook environment for approved users.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: JupyterHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Code-first notebook environment for approved users. Model metadata: source ID C-JUPYTERHUB; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: JupyterHub chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.23. Keycloak

Internal app in the The PaM Data Mesh domain.

Status: live

OIDC identity provider; local users in prototype and LDAP federation in production.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Keycloak chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

OIDC identity provider; local users in prototype and LDAP federation in production. Model metadata: source ID C-KEYCLOAK; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Keycloak chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.24. Keycloak

Internal app in the dlh-in-a-box domain.

Status: live

Optional in-chart OIDC identity provider that issues browser and service authentication tokens and supplies role or group claims.

Locations: icddr,b

Technologies:

- [Keycloak](#)

Links:

- [GitHub: Keycloak chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Identity And Secrets](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional in-chart OIDC identity provider that issues browser and service authentication tokens and supplies role or group claims.

Model metadata: source ID DLH-R-KEYCLOAK; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Identity And Secrets. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Keycloak chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.25. Kubernetes API / Runtime

Internal app in the The PaM Data Mesh domain.

Status: live

Kubernetes API and runtime surface for apps and stores.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Kubernetes configuration](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Kubernetes API and runtime surface for apps and stores. Model metadata: source ID C-K8S; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Kubernetes configuration. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.26. Kubespray Automation

Internal app in the The PaM Data Mesh domain.

Status: live

Bootstraps Kubernetes clusters from inventory and add-on configuration.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: k8s/kubespray](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Bootstraps Kubernetes clusters from inventory and add-on configuration. Model metadata: source ID C-KUBESPRAY; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: k8s/kubespray. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.27. Metrics Server

Internal app in the The PaM Data Mesh domain.

Status: live

Resource metrics for diagnostics and UI tooling.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Metrics Server addon](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Resource metrics for diagnostics and UI tooling. Model metadata: source ID C-METRICS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Metrics Server addon. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.28. OpenStack Cinder CSI

Internal app in the The PaM Data Mesh domain.

Status: live

Dynamic block volume provider for the Sanger prototype cluster.

Locations: Wellcome Sanger Institute

Links:

- [GitHub: development storage configuration](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Dynamic block volume provider for the Sanger prototype cluster. Model metadata: source ID C-CINDER-CSI; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: development storage configuration. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.29. Platform Home

Internal app in the The PaM Data Mesh domain.

Status: live

Portal launcher and access-control UI.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Platform Home template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)
- [Platform Home Component Diagram](#)

Portal launcher and access-control UI. Model metadata: source ID C-PLATFORM-HOME; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Platform Home template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.30. Platform Home

Internal app in the dlh-in-a-box domain.

Status: live

Browser portal that presents authenticated launch links and health information for deployed platform services; the current implementation assumes bundled Keycloak.

Locations: icddr,b

Technologies:

- [HTML](#)

Links:

- [GitHub: Platform Home template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Browser Entry](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Browser portal that presents authenticated launch links and health information for deployed platform services; the current implementation assumes bundled Keycloak.

Model metadata: source ID DLH-R-PLATFORM-HOME; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Browser Entry. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Platform Home template. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.31. Prefect Auth Proxy

Internal app in the The PaM Data Mesh domain.

Status: live

Browser OIDC gate for Prefect UI.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: oauth2-proxy chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Browser OIDC gate for Prefect UI. Model metadata: source ID C-PREFECT-PROXY; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: oauth2-proxy chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.32. Prefect Job Pods

Internal app in the dlh-in-a-box domain.

Status: live

Technologies:

- [Prefect](#)

7.33. Prefect Server

Internal app in the The PaM Data Mesh domain.

Status: live

Workflow orchestration API and UI.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Prefect Server chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Workflow orchestration API and UI. Model metadata: source ID C-PREFECT-SERVER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Prefect Server chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.34. Prefect Server

Internal app in the dlh-in-a-box domain.

Status: live

Orchestration service that stores flow definitions, schedules, run state, and exposes the operational API and UI for data workflows.

Locations: icddr,b

Technologies:

- [Prefect](#)

Links:

- [GitHub: Prefect Server chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Orchestration And Compute](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Orchestration service that stores flow definitions, schedules, run state, and exposes the operational API and UI for data workflows.

Model metadata: source ID DLH-R-PREFECT-SERVER; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Orchestration And Compute. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Prefect Server chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.35. Prefect Worker

Internal app in the dlh-in-a-box domain.

Status: live

Kubernetes-based execution worker that retrieves approved flow runs from Prefect Server and launches the corresponding pipeline jobs.

Locations: icddr,b

Technologies:

- [Prefect](#)

Links:

- [GitHub: Prefect Worker chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Orchestration And Compute](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Kubernetes-based execution worker that retrieves approved flow runs from Prefect Server and launches the corresponding pipeline jobs.

Model metadata: source ID DLH-R-PREFECT-WORKER; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Orchestration And Compute. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Prefect Worker chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.36. Prefect Worker

Internal app in the The PaM Data Mesh domain.

Status: live

Executes flow runs and submits Kubernetes/Spark work.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Prefect Worker chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Executes flow runs and submits Kubernetes/Spark work. Model metadata: source ID C-PREFECT-WORKER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Prefect Worker chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.37. Ranger

Internal app in the dlh-in-a-box domain.

Status: live

Optional policy administration service for governed SQL access, including roles, data masking, row filters, and audit.

Locations: icddr,b

Technologies:

- [Apache Ranger](#)

Links:

- [GitHub: Ranger admin template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Governance](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional policy administration service for governed SQL access, including roles, data masking, row filters, and audit.

Model metadata: source ID DLH-R-RANGER; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Governance. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Ranger admin template. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.38. Ranger Auth / Browser Proxy

Internal app in the The PaM Data Mesh domain.

Status: live

Browser access control and path compatibility for Ranger Admin.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Ranger browser proxy template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Browser access control and path compatibility for Ranger Admin. Model metadata: source ID C-RANGER-PROXY; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Ranger browser proxy template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.39. Ranger Automation Jobs

Internal app in the dlh-in-a-box domain.

Status: live

Technologies:

- [Kubernetes Cron Job](#)

7.40. Render And Validation Automation

Internal app in the The PaM Data Mesh domain.

Status: live

Renders manifests, deploys releases, and validates deployment rules.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Renders manifests, deploys releases, and validates deployment rules. Model metadata: source ID C-DEPLOY-SCRIPTS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.41. Sanger Bastion Host

Internal app in the The PaM Data Mesh domain.

Status: live

SSH jump host for development administration.

Locations: Wellcome Sanger Institute

Links:

- [GitHub: bastion Terraform module](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

SSH jump host for development administration. Model metadata: source ID C-DEV-BASTION; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: bastion Terraform module. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.42. Sanger Kubernetes API

Internal app in the The PaM Data Mesh domain.

Status: live

Development cluster control-plane API.

Locations: Wellcome Sanger Institute

Links:

- [GitHub: development cluster inventory](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In diagrams:

- [Sanger Development Prototype Kubernetes Cluster App Diagram](#)

Development cluster control-plane API. Model metadata: source ID C-DEV-K8S-API; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: development cluster inventory. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.43. Spark Job Pods

Internal app in the dlh-in-a-box domain.

Status: live

Technologies:

- [Apache Spark](#)

7.44. Spark Operator

Internal app in the The PaM Data Mesh domain.

Status: live

Watches SparkApplication CRDs and manages Spark pods.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Spark Operator chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Watches SparkApplication CRDs and manages Spark pods. Model metadata: source ID C-SPARK-OP; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or

hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Spark Operator chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.45. Spark Operator

Internal app in the dlh-in-a-box domain.

Status: live

Kubernetes operator that manages Spark applications submitted by pipeline jobs for distributed processing.

Locations: icddr,b

Technologies:

- [Apache Spark](#)

Links:

- [GitHub: Spark Operator chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Orchestration And Compute](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Kubernetes operator that manages Spark applications submitted by pipeline jobs for distributed processing.

Model metadata: source ID DLH-R-SPARK-OPERATOR; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Orchestration And Compute. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Spark Operator chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.46. Superset

Internal app in the dlh-in-a-box domain.

Status: live

Optional web application for business intelligence, dashboard development, and visual exploration of governed datasets through Trino.

Locations: icddr,b

Technologies:

- [Apache Superset](#)

Links:

- [GitHub: Superset chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Analysis Tools](#)
- [OIDC-enabled browser applications](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional web application for business intelligence, dashboard development, and visual exploration of governed datasets through Trino.

Model metadata: source ID DLH-R-SUPERSET; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Analysis Tools. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Superset chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.47. Superset Worker

Internal app in the The PaM Data Mesh domain.

Status: live

Async query and reporting worker.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Superset chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Async query and reporting worker. Model metadata: source ID C-SUPERSET-WORKER; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Superset chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.48. Terraform Kubernetes Access Automation

Internal app in the The PaM Data Mesh domain.

Status: live

Provisions service accounts, RBAC, OIDC bindings, and kubeconfig outputs.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: terraform](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Provisions service accounts, RBAC, OIDC bindings, and kubeconfig outputs. Model metadata: source ID C-TF-K8S-ACCESS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: terraform. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.49. Terraform OpenStack Automation

Internal app in the The PaM Data Mesh domain.

Status: live

Provisions Sanger prototype infrastructure.

Locations: Wellcome Sanger Institute

Links:

- [GitHub: terraform/openstack](#)

Teams:

- Sanger PaM Data Engineering and Integration
- Sanger IDS (ISG)

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Provisions Sanger prototype infrastructure. Model metadata: source ID C-TF-OPENSTACK; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, Sanger IDS (ISG); location or hosting context Wellcome Sanger Institute; reality link GitHub: terraform/openstack. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.50. Trino

Internal app in the dlh-in-a-box domain.

Status: live

Distributed SQL query engine that provides the principal compute interface for querying and transforming governed lakehouse datasets.

Locations: icddr,b

Technologies:

- [Trino](#)

Links:

- [GitHub: vendored Trino chart](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Lakehouse Core](#)
- [OIDC-enabled browser applications](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Distributed SQL query engine that provides the principal compute interface for querying and transforming governed lakehouse datasets.

Model metadata: source ID DLH-R-TRINO; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Lakehouse Core. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: vendored Trino chart. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.51. Trino Coordinator

Internal app in the The PaM Data Mesh domain.

Status: live

SQL query entry point and Ranger-authorized catalog access.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Trino coordinator template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

SQL query entry point and Ranger-authorized catalog access. Model metadata: source ID C-TRINO; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Trino coordinator template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.52. Trino Workers

Internal app in the The PaM Data Mesh domain.

Status: live

Distributed query execution over lakehouse object data.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Trino worker template](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Distributed query execution over lakehouse object data. Model metadata: source ID C-TRINO-WORKERS; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Trino worker template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.53. User Notebook Pods

Internal app in the dlh-in-a-box domain.

Status: live

Ephemeral Kubernetes pods spawned by JupyterHub to provide isolated notebook runtimes for individual users.

Locations: icddr,b

Technologies:

- [Jupyter](#)

Links:

- [GitHub: JupyterHub chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Analysis Tools](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Ephemeral Kubernetes pods spawned by JupyterHub to provide isolated notebook runtimes for individual users.

Model metadata: source ID DLH-R-JUPYTER-PODS; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Analysis Tools. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: JupyterHub chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.54. Vault

Internal app in the The PaM Data Mesh domain.

Status: live

Runtime service for platform secrets and Kubernetes/OIDC auth.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Vault automation](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)
- [Vault Component Diagram](#)

Runtime service for platform secrets and Kubernetes/OIDC auth. Model metadata: source ID C-VAULT; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Vault automation. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.55. Vault

Internal app in the dlh-in-a-box domain.

Status: live

Optional secret-management service included with the chart; deployments may instead integrate with an external Vault or equivalent secret provider.

Locations: icddr,b

Technologies:

- [HashiCorp Vault](#)

Links:

- [GitHub: Vault chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Identity And Secrets](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional secret-management service included with the chart; deployments may instead integrate with an external Vault or equivalent secret provider.

Model metadata: source ID DLH-R-VAULT; IcePanel type application or executable service; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Identity And Secrets. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Vault chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

7.56. Vault Automation

Internal app in the The PaM Data Mesh domain.

Status: live

Bootstraps Vault, seeds secrets, and syncs runtime secrets.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](https://github.com/platform/dlh-in-a-box/scripts)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Bootstraps Vault, seeds secrets, and syncs runtime secrets. Model metadata: source ID C-VAULT-AUTOMATION; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.57. Vault Secrets Operator

Internal app in the The PaM Data Mesh domain.

Status: live

Syncs VaultStaticSecret resources into Kubernetes Secrets.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Vault Secrets Operator addon](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Syncs VaultStaticSecret resources into Kubernetes Secrets. Model metadata: source ID C-VSO; IcePanel type application or executable service; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Vault Secrets Operator addon. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

7.58. vSphere CSI

Internal app in the The PaM Data Mesh domain.

Status: future

Dynamic VMware persistent disk provider for production PVCs.

Locations: icddr,b

Links:

- [GitHub: production storage configuration](#)

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

Dynamic VMware persistent disk provider for production PVCs. Model metadata: source ID C-VSPHERE-CSI; IcePanel type application or executable service; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production storage configuration. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8. Stores

8.1. Chart Source

Internal store in the dlh-in-a-box domain.

Status: live

Primary source tree for the umbrella Helm chart; defines chart metadata, default configuration, templates, and dependencies.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Chart Source Boundary](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Chart Source Components](#)

Primary source tree for the umbrella Helm chart; defines chart metadata, default configuration, templates, and dependencies.

Model metadata: source ID DLH-C2-CHART-SOURCE; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Chart Source Boundary. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.2. DataHub Kafka

Internal store in the The PaM Data Mesh domain.

Status: live

Metadata event log.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR Data Platform Runtime App Diagram](#)

Metadata event log. Model metadata: source ID C-DATAHUB-KAFKA; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.3. DataHub Kafka

Internal store in the dlh-in-a-box domain.

Status: live

Event-stream service used by DataHub to exchange metadata change events between services.

Locations: icddr,b

Technologies:

- [Apache Kafka](#)

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Event-stream service used by DataHub to exchange metadata change events between services.

Model metadata: source ID DLH-R-DATAHUB-KAFKA; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.4. DataHub MySQL

Internal store in the The PaM Data Mesh domain.

Status: live

DataHub graph and metadata persistence.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

DataHub graph and metadata persistence. Model metadata: source ID C-DATAHUB-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context

Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.5. DataHub MySQL Database

Internal store in the dlh-in-a-box domain.

Status: live

Relational database used by DataHub to persist metadata state and application records.

Locations: icddr,b

Technologies:

- [MySQL](#)

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Relational database used by DataHub to persist metadata state and application records.

Model metadata: source ID DLH-R-DATAHUB-MYSQL; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.6. DataHub Search

Internal store in the The PaM Data Mesh domain.

Status: live

Search index for DataHub.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR Data Platform Runtime App Diagram](#)

Search index for DataHub. Model metadata: source ID C-DATAHUB-SEARCH; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.7. DataHub Search Index

Internal store in the dlh-in-a-box domain.

Status: live

Search index used by DataHub to make metadata entities discoverable in the catalog interface.

Locations: icddr,b

Technologies:

- [Elastic Search](#)

Links:

- [GitHub: DataHub prerequisites chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Search index used by DataHub to make metadata entities discoverable in the catalog interface.

Model metadata: source ID DLH-R-DATAHUB-ELASTICSEARCH; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: DataHub prerequisites chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.8. Environment Settings

Internal store in the The PaM Data Mesh domain.

Status: live

Environment-specific configuration files used when rendering the chart.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/values-dev.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [icddrb-data-platform-infra App Diagram](#)

Environment-specific configuration files used when rendering the chart. Model metadata: source ID C-INFRA-VALUES; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/values-dev.yaml. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.9. Example Values Profiles

Internal store in the dlh-in-a-box domain.

Status: live

Example configuration files showing supported installation modes.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: examples](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Install Profiles](#)

In diagrams:

- [Chart Product And Packaging](#)

Example configuration files that demonstrate and test supported installation modes, including local, development, production, shared identity, in-cluster MinIO, and external S3 configurations.

Model metadata: source ID DLH-C2-EXAMPLE-PROFILES; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Install Profiles. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and

Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: examples. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.10. Helm Chart Lock

Internal store in the The PaM Data Mesh domain.

Status: live

Chart.lock: Resolved dependency versions and digests generated from the metadata dependency declarations

8.11. Helm Chart Metadata

Internal store in the The PaM Data Mesh domain.

Status: live

Chart.yaml: Chart name, version, app version, annotations, dependency declarations, repositories, and conditions

8.12. Helm Chart Source

Internal store in the The PaM Data Mesh domain.

Status: live

charts/dlh-in-a-box: The umbrella helm chart source directory in the repository

Technologies:

- [Folder](#)

8.13. Helm Chart Templates

Internal store in the The PaM Data Mesh domain.

Status: live

templates/: Repo-owned templates for identity, governance, Ranger, CloudBeaver, DataHub, Platform Home & compatibility

8.14. Helm Chart Values Contract

Internal store in the The PaM Data Mesh domain.

Status: live

values.yaml / values.schema.json: Default values, schema, and supported configuration surface exposed to consumers

8.15. Hive Database

Internal store in the dlh-in-a-box domain.

Status: live

PostgreSQL persistence layer for Hive Metastore catalog metadata.

Locations: icddr,b

Technologies:

- [PostgreSQL](#)

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

PostgreSQL persistence layer for Hive Metastore catalog metadata.

Model metadata: source ID DLH-R-HIVE-DB; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.16. Hive Metastore Local Subchart

Internal store in the The PaM Data Mesh domain.

Status: live

Technologies:

- [Helm](#)

In groups:

- [Local File Dependency Charts](#)

8.17. Hive Metastore Local Subchart

Internal store in the dlh-in-a-box domain.

Status: live

First-party Helm subchart that renders the optional Hive Metastore service used for SQL catalog metadata.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive](#)
- [GitHub: charts/dlh-in-a-box/charts/hive-0.1.0.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Chart Source Boundary](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Hive Metastore Local Subchart](#)

First-party Helm subchart that renders the optional Hive Metastore service used for SQL catalog metadata.

Model metadata: source ID DLH-C2-HIVE-SUBCHART; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Chart Source Boundary. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/hive; GitHub: charts/dlh-in-a-box/charts/hive-0.1.0.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.18. Hive PostgreSQL

Internal store in the The PaM Data Mesh domain.

Status: live

Hive metastore database.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Hive metastore database. Model metadata: source ID C-HIVE-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.19. In-Cluster Object Store

Internal store in the dlh-in-a-box domain.

Status: live

Optional S3-compatible object store deployed inside the cluster when a deployment does not use an external object-storage service.

Locations: icddr,b

Technologies:

- [MinIO](#)

Links:

- [GitHub: MinIO chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Lakehouse Core](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Optional S3-compatible object store deployed inside the cluster when a deployment does not use an external object-storage service.

Model metadata: source ID DLH-R-MINIO; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Lakehouse Core. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: MinIO chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.20. Keycloak Database

Internal store in the dlh-in-a-box domain.

Status: live

PostgreSQL persistence layer for Keycloak realms, clients, users, sessions, and configuration when bundled Keycloak is enabled.

Locations: icddr,b

Technologies:

- [PostgreSQL](#)

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

PostgreSQL persistence layer for Keycloak realms, clients, users, sessions, and configuration when bundled Keycloak is enabled.

Model metadata: source ID DLH-R-KEYCLOAK-DB; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.21. Keycloak PostgreSQL

Internal store in the The PaM Data Mesh domain.

Status: live

Keycloak realm, user, and session store.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRb Data Platform Runtime App Diagram](#)

Keycloak realm, user, and session store. Model metadata: source ID C-KEYCLOAK-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.22. MinIO

Internal store in the The PaM Data Mesh domain.

Status: live

Lakehouse object store for platform-managed data.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: MinIO chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDRB Data Platform Runtime App Diagram](#)

Lakehouse object store for platform-managed data. Model metadata: source ID C-MINIO; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: MinIO chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddr-b-data-platform-infra repository.

8.23. Packaged Upstream Dependency Helm Charts

Internal store in the The PaM Data Mesh domain.

Status: live

Superset, Prefect Server, Prefect Worker, OAuth2 Proxy, KeyCloak, Spark, Minio, Datahub, Vault, Superset, PostgreSQL

Technologies:

- [Tar Gzip](#)

8.24. Packaged Upstream Dependency Helm Charts

Internal store in the dlh-in-a-box domain.

Status: live

Version-resolved Helm dependency archives bundled with the chart to make dependency resolution explicit and reproducible.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Chart Source Boundary](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Packaged Upstream Dependencies](#)

Version-resolved Helm dependency archives bundled with the chart to make dependency resolution explicit and reproducible.

Model metadata: source ID DLH-C2-UPSTREAM-ARCHIVES; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Chart Source Boundary. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.25. Prefect Database

Internal store in the dlh-in-a-box domain.

Status: live

PostgreSQL store that persists Prefect orchestration metadata, including flow runs, schedules, and state transitions.

Locations: icddr,b

Technologies:

- [PostgreSQL](#)

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

PostgreSQL store that persists Prefect orchestration metadata, including flow runs, schedules, and state transitions.

Model metadata: source ID DLH-R-PREFECT-DB; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.26. Prefect PostgreSQL

Internal store in the The PaM Data Mesh domain.

Status: live

Prefect state database.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Prefect state database. Model metadata: source ID C-PREFECT-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.27. Published OCI Chart Package

Internal store in the dlh-in-a-box domain.

Status: live

Versioned Helm chart artifact stored in the GHCR OCI registry and consumed by deployment repositories.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: chart package metadata](#)
- [GitHub: chart publish workflow](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Published Artifact](#)

In diagrams:

- [Chart Product And Packaging](#)

Versioned Helm chart artifact stored in the GHCR OCI registry and consumed by deployment repositories.

Model metadata: source ID DLH-C2-OCI-PACKAGE; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Published Artifact. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: chart package metadata; GitHub: chart publish workflow. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.28. Ranger Database

Internal store in the dlh-in-a-box domain.

Status: live

PostgreSQL store for Ranger policy definitions, users, roles, service metadata, and audit-related state.

Locations: icddr,b

Technologies:

- [PostgreSQL](#)

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

PostgreSQL store for Ranger policy definitions, users, roles, service metadata, and audit-related state.

Model metadata: source ID DLH-R-RANGER-DB; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.29. Ranger PostgreSQL

Internal store in the The PaM Data Mesh domain.

Status: live

Ranger policy and audit persistence.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Ranger policy and audit persistence. Model metadata: source ID C-RANGER-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.30. Shared Framework Package

Internal store in the The PaM Data Mesh domain.

Status: future

Target package boundary for common data-pipeline behaviours.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: Sanger Data Lakehouse packages](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)
- [Shared Python Data Pipelines Framework App Diagram](#)

Target package boundary for common data-pipeline behaviours. Model metadata: source ID C-FRAMEWORK-PACKAGE; IcePanel type data store or repository; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location

or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: Sanger Data Lakehouse packages. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.31. Superset Database

Internal store in the dlh-in-a-box domain.

Status: live

Relational metadata store for Superset users, dashboards, charts, datasets, and application settings.

Locations: icddr,b

Technologies:

- [PostgreSQL](#)

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Relational metadata store for Superset users, dashboards, charts, datasets, and application settings.

Model metadata: source ID DLH-R-SUPERSET-DB; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.32. Superset PostgreSQL

Internal store in the The PaM Data Mesh domain.

Status: live

Superset metadata store.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: PostgreSQL chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDR B Data Platform Runtime App Diagram](#)

Superset metadata store. Model metadata: source ID C-SUPERSET-DB; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: PostgreSQL chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.33. Superset Queue

Internal store in the dlh-in-a-box domain.

Status: live

Cache and asynchronous task queue used by Superset for responsive dashboard execution and background work.

Locations: icddr,b

Technologies:

- [Redis](#)

Links:

- [GitHub: Superset chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In groups:

- [Support Services and Persistent Data Stores](#)
- [Support Services And Stores](#)

In diagrams:

- [Deployed DLH-in-a-box Runtime App Diagram](#)

Cache and asynchronous task queue used by Superset for responsive dashboard execution and background work.

Model metadata: source ID DLH-R-SUPERSET-REDIS; IcePanel type data store or repository; Child of Deployed DLH-in-a-box Runtime (DLH-L1-RUNTIME). IcePanel group membership: Support Services And Stores. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration, icddr,b IT. Location or hosting context: icddr,b. Reality links: GitHub: Superset chart archive. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.34. Superset Redis

Internal store in the The PaM Data Mesh domain.

Status: live

Celery broker and cache.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Superset chart archive](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [ICDDRB Data Platform Runtime App Diagram](#)

Celery broker and cache. Model metadata: source ID C-SUPERSET-REDIS; IcePanel type data store or repository; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Superset chart archive. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

8.35. Vendored Trino Chart

Internal store in the dlh-in-a-box domain.

Status: live

Local copy of the upstream Trino chart adapted for DLH-in-a-box identity, storage, and access-control integration.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino](#)
- [GitHub: charts/dlh-in-a-box/charts/trino-0.34.0.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In groups:

- [Chart Source Boundary](#)

In diagrams:

- [Chart Product And Packaging](#)
- [Vendored Trino Chart](#)

Local copy of the upstream Trino chart adapted for DLH-in-a-box identity, storage, and access-control integration.

Model metadata: source ID DLH-C2-TRINO-VENDORED; IcePanel type data store or repository; Child of DLH-in-a-box Umbrella Helm Chart (DLH-L1-CHART). IcePanel group membership: Chart Source Boundary. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/trino; GitHub: charts/dlh-in-a-box/charts/trino-0.34.0.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

8.36. Vendored Trino Helm Chart

Internal store in the The PaM Data Mesh domain.

Status: live

charts/dlh-in-a-box/charts/trino/; charts/dlh-in-a-box/charts/trino-0.34.0.tgz: Vendored Trino chart source with patches

Technologies:

- [Helm](#)

In groups:

- [Local File Dependency Charts](#)

This repo has local patch points for catalog generation, object storage, identity/OIDC, access control, and coordinator/worker wiring

8.37. VMware Persistent Disks

Internal store in the The PaM Data Mesh domain.

Status: future

VMware-backed block storage for production services.

Locations: icddr,b

Links:

- [GitHub: production storage architecture](#)

Teams:

- icddr,b IT
- Sanger PaM Data Engineering and Integration

In diagrams:

- [icddr,b Production Kubernetes Cluster App Diagram](#)

VMware-backed block storage for production services. Model metadata: source ID C-PROD-VMWARE-DISKS; IcePanel type data store or repository; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship icddr,b IT, Sanger PaM Data Engineering and Integration; location or hosting context icddr,b; reality link GitHub: production storage architecture. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9. Components

9.1. Access Control API

Internal component in the The PaM Data Mesh domain.

Status: live

Resolves app visibility from user groups and cached Ranger/platform role state.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Platform Home template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Resolves app visibility from user groups and cached Ranger/platform role state. Model metadata: source ID CMP-HOME-API; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Platform Home template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.2. Access Control ConfigMap

Internal component in the The PaM Data Mesh domain.

Status: live

Cached group-to-service role state used by portal rendering.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Platform Home template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Cached group-to-service role state used by portal rendering. Model metadata: source ID CMP-HOME-CM; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Platform Home template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.3. Access Control Reconcile CronJob

Internal component in the The PaM Data Mesh domain.

Status: live

Refreshes access-control cache from Ranger.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Ranger automation template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Refreshes access-control cache from Ranger. Model metadata: source ID CMP-HOME-RECONCILE; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Ranger automation template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.4. blocks

Internal component in the The PaM Data Mesh domain.

Status: future

Block specs, builders, validation, and CLI helpers.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-flow-utils package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Block specs, builders, validation, and CLI helpers. Model metadata: source ID CMP-FW-BLOCKS; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-flow-utils package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.5. Bronze Zone

Internal component in the dlh-in-a-box domain.

Status: live

Raw Data

9.6. Chart Guide And Notices

Internal component in the dlh-in-a-box domain.

Status: live

Provides chart-local operating guidance, licensing information, and third-party notices for redistributed material.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/README.md](#)
- [GitHub: charts/dlh-in-a-box/LICENSE](#)
- [GitHub: charts/dlh-in-a-box/THIRD_PARTY_NOTICES.md](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Provides chart-local operating guidance, licensing information, and third-party notices for redistributed material.

Model metadata: source ID DLH-C3-SOURCE-DOCS; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/README.md; GitHub: charts/dlh-in-a-box/LICENSE; GitHub: charts/dlh-in-a-box/THIRD_PARTY_NOTICES.md. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.7. Chart Lock

Internal component in the dlh-in-a-box domain.

Status: live

Records exact dependency versions and digests, providing reproducible dependency resolution.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/Chart.lock](https://github.com/wellcome-lab/dlh-in-a-box/Chart.lock)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Records exact dependency versions and digests, providing reproducible dependency resolution.

Model metadata: source ID DLH-C3-SOURCE-LOCK; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/Chart.lock. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.8. Chart Metadata

Internal component in the dlh-in-a-box domain.

Status: live

Declares the chart identity and its Helm dependencies, component names, repositories, and activation conditions.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/Chart.yaml](https://github.com/wellcome-lab/dlh-in-a-box/Chart.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Declares the chart identity and its Helm dependencies, component names, repositories, and activation conditions.

Model metadata: source ID DLH-C3-SOURCE-METADATA; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/Chart.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.9. Chart Metadata

Internal component in the The PaM Data Mesh domain.

Status: live

Chart identity, versioning, annotations, sources, maintainers, dependency declarations, aliases, repositories, conditions

Technologies:

- [YAML](#)

9.10. CLI

Internal component in the The PaM Data Mesh domain.

Status: future

Common command surface for blocks, deployments, metadata, and developer ergonomics.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-flow-utils package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Common command surface for blocks, deployments, metadata, and developer ergonomics. Model metadata: source ID CMP-FW-CLI; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-flow-utils package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.11. CloudBeaver Templates

Internal component in the dlh-in-a-box domain.

Status: live

Configure CloudBeaver workspace initialization, Trino connection definitions, user permissions, and authentication proxy integration.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/cloudbeaver.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Configure CloudBeaver workspace initialization, Trino connection definitions, user permissions, and authentication proxy integration.

Model metadata: source ID DLH-C3-SOURCE-CLOUDBEAVER; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/cloudbeaver.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.12. connectors

Internal component in the The PaM Data Mesh domain.

Status: future

Base connector interfaces and reusable source-family modules.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-data-sources package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Base connector interfaces and reusable source-family modules. Model metadata: source ID CMP-FW-CONNECTORS; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-data-sources package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.13. DataHub Auth And Compat Templates

Internal component in the dlh-in-a-box domain.

Status: live

Provide DataHub authentication secrets and compatibility resources for optional DataHub prerequisite services.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/datahub-auth-secrets.yaml](#)
- [GitHub: charts/dlh-in-a-box/templates/datahub-prerequisites-compat.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Provide DataHub authentication secrets and compatibility resources for optional DataHub prerequisite services.

Model metadata: source ID DLH-C3-SOURCE-DATAHUB; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/datahub-auth-secrets.yaml; GitHub: charts/dlh-in-a-box/templates/datahub-prerequisites-compat.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.14. DataHub Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for optional DataHub metadata catalog and discovery services.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/datahub-0.8.21.tgz](https://github.com/wellcome-sanger/dlh-in-a-box/charts/datahub-0.8.21.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for optional DataHub metadata catalog and discovery services.

Model metadata: source ID DLH-C3-PKG-DATAHUB; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/datahub-0.8.21.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.15. DataHub Prerequisites Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for optional DataHub support services, including persistence, messaging, and search dependencies.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/datahub-prerequisites-0.2.3.tgz](https://github.com/charts/dlh-in-a-box/charts/datahub-prerequisites-0.2.3.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for optional DataHub support services, including persistence, messaging, and search dependencies.

Model metadata: source ID DLH-C3-PKG-DATAHUB-PREREQS; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/datahub-prerequisites-0.2.3.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.16. dbt helpers

Internal component in the The PaM Data Mesh domain.

Status: future

Profile wiring, selectors, and common dbt execution helpers.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-flow-utils package](https://gitlab.com/dlh-flow-utils/package)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Profile wiring, selectors, and common dbt execution helpers. Model metadata: source ID CMP-FW-DBT; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or

hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-flow-utils package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.17. Dependency Refresh Check

Internal component in the dlh-in-a-box domain.

Status: live

Refreshes Helm dependency archives and updates lock metadata so packaged dependencies remain reproducible.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: hack/helm-dependency-update.sh](#)
- [GitHub: charts/dlh-in-a-box/Chart.lock](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Validation Automation](#)

Refreshes Helm dependency archives and updates lock metadata so packaged dependencies remain reproducible.

Model metadata: source ID DLH-C3-VALIDATE-DEPS; IcePanel type component within a parent object; Child of Chart Validation Automation (DLH-C2-VALIDATION). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: hack/helm-dependency-update.sh; GitHub: charts/dlh-in-a-box/Chart.lock. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.18. deployments

Internal component in the The PaM Data Mesh domain.

Status: future

Deployment spec models, registry, validation, and image resolution.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-project-template](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Deployment spec models, registry, validation, and image resolution. Model metadata: source ID CMP-FW-DEPLOY; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-project-template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.19. Example Render Checks

Internal component in the dlh-in-a-box domain.

Status: live

Renders every maintained example configuration file to verify that supported installation profiles produce valid manifests.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: hack/template.sh](#)
- [GitHub: examples](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Validation Automation](#)

Renders every maintained example configuration file to verify that supported installation profiles produce valid manifests.

Model metadata: source ID DLH-C3-VALIDATE-RENDER; IcePanel type component within a parent object; Child of Chart Validation Automation (DLH-C2-VALIDATION). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [hack/template.sh](#); GitHub: [examples](#). Source of truth: [docs/architecture/icepanel/models/dlh-in-a-box.json](#) in the DLH-in-a-box chart repository.

9.20. Gold Zone

Internal component in the dlh-in-a-box domain.

Status: live

Curated, Aggregated Analysis Ready Data

9.21. Governance Validation Templates

Internal component in the dlh-in-a-box domain.

Status: live

Validate authorization settings, including roles, Ranger integration, catalog definitions, and policy settings.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/governance-validation.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Validate authorization settings, including roles, Ranger integration, catalog definitions, and policy settings.

Model metadata: source ID DLH-C3-SOURCE-GOVERNANCE; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/templates/governance-validation.yaml](#). Source of truth: [docs/architecture/icepanel/models/dlh-in-a-box.json](#) in the DLH-in-a-box chart repository.

9.22. Helm Lint Workflow

Internal component in the dlh-in-a-box domain.

Status: live

Runs automated checks for licensing, documentation, security, rendering rules, shell scripts, JSON schema, and Helm syntax.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/workflows/helm-lint.yaml](#)
- [GitHub: hack/lint.sh](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Validation Automation](#)

Runs automated checks for licensing, documentation, security, rendering rules, shell scripts, JSON schema, and Helm syntax.

Model metadata: source ID DLH-C3-VALIDATE-LINT; IcePanel type component within a parent object; Child of Chart Validation Automation (DLH-C2-VALIDATION). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/workflows/helm-lint.yaml; GitHub: hack/lint.sh. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.23. Helm Package Step

Internal component in the dlh-in-a-box domain.

Status: live

Builds the versioned Helm chart archive for release.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: hack/package.sh](https://github.com/hack/package.sh)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Publish Automation](#)

Builds the versioned Helm chart archive for release.

Model metadata: source ID DLH-C3-PUBLISH-PACKAGE; IcePanel type component within a parent object; Child of Chart Publish Automation (DLH-C2-PUBLISH). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: hack/package.sh. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.24. Hive Chart Metadata

Internal component in the dlh-in-a-box domain.

Status: live

Declares the local subchart identity, version, and chart metadata.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/Chart.yaml](https://github.com/charts/dlh-in-a-box/charts/hive/Chart.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Declares the local subchart identity, version, and chart metadata.

Model metadata: source ID DLH-C3-HIVE-METADATA; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering

and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/hive/Chart.yaml](#). Source of truth: [docs/architecture/icepanel/models/dlh-in-a-box.json](#) in the DLH-in-a-box chart repository.

9.25. Hive Config Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Generates Hive Metastore settings for database connectivity and object-storage access.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/templates/configmap.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Generates Hive Metastore settings for database connectivity and object-storage access.

Model metadata: source ID DLH-C3-HIVE-CONFIG; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/hive/templates/configmap.yaml](#). Source of truth: [docs/architecture/icepanel/models/dlh-in-a-box.json](#) in the DLH-in-a-box chart repository.

9.26. Hive Helpers

Internal component in the dlh-in-a-box domain.

Status: live

Provides template helpers for naming, secret lookup, S3 settings, PostgreSQL settings, and checksum annotations.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/templates/_helpers.tpl](https://github.com/wellcome-lab/dlh-in-a-box/charts/hive/templates/_helpers.tpl)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Provides template helpers for naming, secret lookup, S3 settings, PostgreSQL settings, and checksum annotations.

Model metadata: source ID DLH-C3-HIVE-HELPERS; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/hive/templates/_helpers.tpl. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.27. Hive Metastore Runtime Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Renders the Kubernetes Service, Deployment, volume mounts, probes, and optional ingress for Hive Metastore.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/templates/metastore.yaml](https://github.com/wellcome-lab/dlh-in-a-box/charts/hive/templates/metastore.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Renders the Kubernetes Service, Deployment, volume mounts, probes, and optional ingress for Hive Metastore.

Model metadata: source ID DLH-C3-HIVE-METASTORE; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration.

and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/hive/templates/metastore.yaml](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.28. Hive Schema Initialization

Internal component in the dlh-in-a-box domain.

Status: live

Initializes or validates the relational schema required by the Hive Metastore database.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/templates/init-config.yaml](#)
- [GitHub: charts/dlh-in-a-box/charts/hive/templates/init-schema-job.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Initializes or validates the relational schema required by the Hive Metastore database.

Model metadata: source ID DLH-C3-HIVE-SCHEMA; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/hive/templates/init-config.yaml](#); GitHub: [charts/dlh-in-a-box/charts/hive/templates/init-schema-job.yaml](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.29. Hive Secret Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Generates Kubernetes Secrets for database and object-storage credentials used by Hive.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/templates/postgres-secret.yaml](https://github.com/dlh-in-a-box/charts/hive/templates/postgres-secret.yaml)
- [GitHub: charts/dlh-in-a-box/charts/hive/templates/s3-secret.yaml](https://github.com/dlh-in-a-box/charts/hive/templates/s3-secret.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Generates Kubernetes Secrets for database and object-storage credentials used by Hive.

Model metadata: source ID DLH-C3-HIVE-SECRETS; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/hive/templates/postgres-secret.yaml; GitHub: charts/dlh-in-a-box/charts/hive/templates/s3-secret.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.30. Hive Subchart Documentation

Internal component in the dlh-in-a-box domain.

Status: live

Defines the purpose and ownership boundary of the first-party Hive Metastore subchart.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/README.md](https://github.com/dlh-in-a-box/charts/hive/README.md)
- [GitHub: charts/dlh-in-a-box/charts/hive/templates/ README.txt](https://github.com/dlh-in-a-box/charts/hive/templates/README.txt)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Defines the purpose and ownership boundary of the first-party Hive Metastore subchart.

Model metadata: source ID DLH-C3-HIVE-DOCS; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/hive/README.md; GitHub: charts/dlh-in-a-box/charts/hive/templates/_README.txt. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.31. Hive Values

Internal component in the dlh-in-a-box domain.

Status: live

Defines Hive-specific defaults, including metastore, database, and object-storage settings.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/hive/values.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Hive Metastore Local Subchart](#)

Defines Hive-specific defaults, including metastore, database, and object-storage settings.

Model metadata: source ID DLH-C3-HIVE-VALUES; IcePanel type component within a parent object; Child of Hive Metastore Local Subchart (DLH-C2-HIVE-SUBCHART). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/hive/values.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.32. Human Access Sync

Internal component in the The PaM Data Mesh domain.

Status: live

Reads development access roster and provisions users, groups, and per-user Vault paths.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/vault/dev-human-access.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Vault Component Diagram](#)

Reads development access roster and provisions users, groups, and per-user Vault paths. Model metadata: source ID CMP-VAULT-HUMAN; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/vault/dev-human-access.yaml. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.33. Identity Validation Templates

Internal component in the dlh-in-a-box domain.

Status: live

Validate identity-provider settings, including OIDC settings, LDAP integration, clients, redirect URIs, and group conventions.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/identity-validation.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Validate identity-provider settings, including OIDC settings, LDAP integration, clients, redirect URIs, and group conventions.

Model metadata: source ID DLH-C3-SOURCE-IDENTITY; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/identity-validation.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.34. JupyterHub Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream JupyterHub service, enabling authenticated multi-user notebook environments.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/jupyterhub-4.3.3.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream JupyterHub service, enabling authenticated multi-user notebook environments.

Model metadata: source ID DLH-C3-PKG-JUPYTERHUB; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/jupyterhub-4.3.3.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.35. Keycloak Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream Keycloak identity provider used when OIDC is included with the deployment.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/keycloak-25.2.0.tgz](https://github.com/dlh-in-a-box/charts/keycloak-25.2.0.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream Keycloak identity provider used when OIDC is included with the deployment.

Model metadata: source ID DLH-C3-PKG-KEYCLOAK; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/keycloak-25.2.0.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.36. Kubernetes Access Sync

Internal component in the The PaM Data Mesh domain.

Status: live

Mirrors Terraform-generated kubeconfigs into Vault.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](https://github.com/platform/dlh-in-a-box/scripts)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Vault Component Diagram](#)

Mirrors Terraform-generated kubeconfigs into Vault. Model metadata: source ID CMP-VAULT-K8S; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.37. metadata

Internal component in the The PaM Data Mesh domain.

Status: future

Ownership, lineage, and metadata publication helpers.

Locations: Wellcome Sanger Institute, Cloud

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Ownership, lineage, and metadata publication helpers. Model metadata: source ID CMP-FW-METADATA; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.38. MinIO Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream MinIO object store used when lakehouse data is stored inside the cluster.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/minio-15.0.7.tgz](https://github.com/dlh-in-a-box/charts/minio-15.0.7.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream MinIO object store used when lakehouse data is stored inside the cluster.

Model metadata: source ID DLH-C3-PKG-MINIO; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/minio-15.0.7.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.39. nginx static server

Internal component in the The PaM Data Mesh domain.

Status: live

Serves portal SPA and static assets.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Platform Home static assets](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Serves portal SPA and static assets. Model metadata: source ID CMP-HOME-NGINX; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Platform Home static assets. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddr-b-data-platform-infra repository.

9.40. oauth2-proxy Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for oauth2-proxy instances that protect browser-facing services with OIDC authentication.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/oauth2-proxy-10.1.4.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for oauth2-proxy instances that protect browser-facing services with OIDC authentication.

Model metadata: source ID DLH-C3-PKG-OAUTH2-PROXY; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/oauth2-proxy-10.1.4.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.41. OCI Push Step

Internal component in the dlh-in-a-box domain.

Status: live

Publishes the release chart package to the GHCR OCI registry.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/workflows/helm-publish.yaml](https://github.com/.github/workflows/helm-publish.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Publish Automation](#)

Publishes the release chart package to the GHCR OCI registry.

Model metadata: source ID DLH-C3-PUBLISH-OCI; IcePanel type component within a parent object; Child of Chart Publish Automation (DLH-C2-PUBLISH). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/workflows/helm-publish.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.42. optional Spark

Internal component in the The PaM Data Mesh domain.

Status: future

Optional Spark integration helpers where a product needs distributed execution.

Locations: Wellcome Sanger Institute, Cloud

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Optional Spark integration helpers where a product needs distributed execution. Model metadata: source ID CMP-FW-SPARK; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and

Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.43. orchestration

Internal component in the The PaM Data Mesh domain.

Status: future

Shared tasks, flow runtime helpers, and optional Spark submission helpers.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-flow-utils package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Shared tasks, flow runtime helpers, and optional Spark submission helpers. Model metadata: source ID CMP-FW-ORCH; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-flow-utils package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.44. Packaging Controls

Internal component in the dlh-in-a-box domain.

Status: live

Specifies files excluded from distributed chart archives, limiting package contents to deployable material.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/.helmignore](https://github.com/charts/dlh-in-a-box/.helmignore)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Specifies files excluded from distributed chart archives, limiting package contents to deployable material.

Model metadata: source ID DLH-C3-SOURCE-PACKAGING; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/.helmignore. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.45. Platform Home Templates

Internal component in the dlh-in-a-box domain.

Status: live

Render the platform portal, launcher settings, runtime settings, and health-check endpoints.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/platform-home.yaml](https://github.com/charts/dlh-in-a-box/templates/platform-home.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Render the platform portal, launcher settings, runtime settings, and health-check endpoints.

Model metadata: source ID DLH-C3-SOURCE-HOME; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location

or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/platform-home.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.46. PostgreSQL Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream PostgreSQL chart reused by service-specific database components.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/postgresql-14.3.3.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream PostgreSQL chart reused by service-specific database components.

Model metadata: source ID DLH-C3-PKG-POSTGRESQL; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/postgresql-14.3.3.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.47. Prefect Server Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream Prefect Server service, enabling workflow orchestration.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/prefect-server-2025.12.31221620.tgz](https://github.com/charts/dlh-in-a-box/charts/prefect-server-2025.12.31221620.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream Prefect Server service, enabling workflow orchestration.

Model metadata: source ID DLH-C3-PKG-PREFECT-SERVER; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/prefect-server-2025.12.31221620.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.48. Prefect Worker Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream Prefect Worker service, enabling Kubernetes-based workflow execution.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/prefect-worker-2025.12.31221620.tgz](https://github.com/charts/dlh-in-a-box/charts/prefect-worker-2025.12.31221620.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream Prefect Worker service, enabling Kubernetes-based workflow execution.

Model metadata: source ID DLH-C3-PKG-PREFECT-WORKER; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger

PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/prefect-worker-2025.12.31221620.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.49. quality

Internal component in the The PaM Data Mesh domain.

Status: future

Reusable data-quality checks, contracts, and monitoring hooks.

Locations: Wellcome Sanger Institute, Cloud

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Reusable data-quality checks, contracts, and monitoring hooks. Model metadata: source ID CMP-FW-QUALITY; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link No Git reality link is assigned; object is conceptual, external, or still awaiting a precise repository target. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.50. Ranger Automation Templates

Internal component in the dlh-in-a-box domain.

Status: live

Generate Kubernetes resources that bootstrap, synchronize, administer, and audit Ranger policy state.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/ranger-automation.yaml](#)
- [GitHub: charts/dlh-in-a-box/templates/ranger-admin.yaml](#)

- [GitHub: charts/dlh-in-a-box/templates/_ranger-admin.tpl](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Generate Kubernetes resources that bootstrap, synchronize, administer, and audit Ranger policy state.

Model metadata: source ID DLH-C3-SOURCE-RANGER; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/ranger-automation.yaml; GitHub: charts/dlh-in-a-box/templates/ranger-admin.yaml; GitHub: charts/dlh-in-a-box/templates/_ranger-admin.tpl. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.51. Ranger Browser Proxy Templates

Internal component in the dlh-in-a-box domain.

Status: live

Render oauth2-proxy resources that protect the Ranger administrative browser interface.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates/ranger-browser-proxy.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Render oauth2-proxy resources that protect the Ranger administrative browser interface.

Model metadata: source ID DLH-C3-SOURCE-RANGER-PROXY; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration.

Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates/ranger-browser-proxy.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.52. Ranger Exception Audit CronJob

Internal component in the The PaM Data Mesh domain.

Status: live

Audits short-lived Ranger exceptions.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Ranger automation template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Audits short-lived Ranger exceptions. Model metadata: source ID CMP-HOME-AUDIT; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Ranger automation template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.53. Ranger Local User Sync CronJob

Internal component in the The PaM Data Mesh domain.

Status: live

Projects Keycloak-local users and groups into Ranger in development.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: Ranger automation template](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Platform Home Component Diagram](#)

Projects Keycloak-local users and groups into Ranger in development. Model metadata: source ID CMP-HOME-USER-SYNC; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: Ranger automation template. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.54. Release Metadata Step

Internal component in the dlh-in-a-box domain.

Status: live

Records chart version, labels, annotations, and release metadata used by deployment repositories.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/release.yml](#)
- [GitHub: charts/dlh-in-a-box/Chart.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Publish Automation](#)

Records chart version, labels, annotations, and release metadata used by deployment repositories.

Model metadata: source ID DLH-C3-PUBLISH-METADATA; IcePanel type component within a parent object; Child of Chart Publish Automation (DLH-C2-PUBLISH). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/release.yml; GitHub: charts/dlh-in-a-box/Chart.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.55. Render Contract Checks

Internal component in the dlh-in-a-box domain.

Status: live

Verifies expected render outputs and expected failure modes for the chart's documented configuration rules.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: hack/render-contract.sh](#)
- [GitHub: hack/testdata/render-contract](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Validation Automation](#)

Verifies expected render outputs and expected failure modes for the chart's documented configuration rules.

Model metadata: source ID DLH-C3-VALIDATE-CONTRACT; IcePanel type component within a parent object; Child of Chart Validation Automation (DLH-C2-VALIDATION). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: hack/render-contract.sh; GitHub: hack/testdata/render-contract. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.56. schema parsing

Internal component in the The PaM Data Mesh domain.

Status: future

Config-driven schema parsing, mappings, and generated view/model support.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-schema-parser package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Config-driven schema parsing, mappings, and generated view/model support. Model metadata: source ID CMP-FW-SCHEMA; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-schema-parser package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.57. Secret Seeding Script

Internal component in the The PaM Data Mesh domain.

Status: live

Writes runtime secrets from local ignored seed files to Vault.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](#)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Vault Component Diagram](#)

Writes runtime secrets from local ignored seed files to Vault. Model metadata: source ID CMP-VAULT-SEED; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.58. settings/config

Internal component in the The PaM Data Mesh domain.

Status: future

Typed config, refs, environment loading, and shared runtime settings.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-schema-parser package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Typed config, refs, environment loading, and shared runtime settings. Model metadata: source ID CMP-FW-SETTINGS; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-schema-parser package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.59. Silver Zone

Internal component in the dlh-in-a-box domain.

Status: live

Cleaned, Deduplicated and Validated Data

9.60. Smoke Install Workflow

Internal component in the dlh-in-a-box domain.

Status: live

Installs an identity- and governance-oriented local profile into a disposable cluster to test an integrated runtime path.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: .github/workflows/helm-smoke-install.yaml](https://github.com/workflows/helm-smoke-install.yaml)
- [GitHub: hack/smoke-install.sh](https://github.com/hack/smoke-install.sh)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Validation Automation](#)

Installs an identity- and governance-oriented local profile into a disposable cluster to test an integrated runtime path.

Model metadata: source ID DLH-C3-VALIDATE-SMOKE; IcePanel type component within a parent object; Child of Chart Validation Automation (DLH-C2-VALIDATION). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: .github/workflows/helm-smoke-install.yaml; GitHub: hack/smoke-install.sh. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.61. Spark Operator Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream Spark Operator, enabling Kubernetes-native management of Spark applications.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/spark-operator-2.4.0.tgz](https://github.com/dlh-in-a-box/charts/spark-operator-2.4.0.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream Spark Operator, enabling Kubernetes-native management of Spark applications.

Model metadata: source ID DLH-C3-PKG-SPARK; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/spark-operator-2.4.0.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.62. Static Payload Files

Internal component in the dlh-in-a-box domain.

Status: live

Contains non-template files copied into generated Kubernetes objects, including Platform Home application assets.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/files](https://github.com/dlh-in-a-box/files)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Contains non-template files copied into generated Kubernetes objects, including Platform Home application assets.

Model metadata: source ID DLH-C3-SOURCE-STATIC-FILES; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration.

Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/files. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.63. storage

Internal component in the The PaM Data Mesh domain.

Status: future

Object-store adapters, Delta/Parquet layout helpers, and lifecycle hooks.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-storage-backends package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Object-store adapters, Delta/Parquet layout helpers, and lifecycle hooks. Model metadata: source ID CMP-FW-STORAGE; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-storage-backends package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.64. Superset Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for the upstream Superset service, enabling optional dashboarding and business-intelligence deployment.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/superset-0.15.2.tgz](https://github.com/dlh-in-a-box/charts/superset-0.15.2.tgz)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for the upstream Superset service, enabling optional dashboarding and business-intelligence deployment.

Model metadata: source ID DLH-C3-PKG-SUPERSET; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/superset-0.15.2.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.65. Third-Party Provenance

Internal component in the dlh-in-a-box domain.

Status: live

Stores bundled third-party license and notice material needed to document redistributed dependencies.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/third_party](https://github.com/dlh-in-a-box/third_party)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Stores bundled third-party license and notice material needed to document redistributed dependencies.

Model metadata: source ID DLH-C3-SOURCE-PROVENANCE; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration.

Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/third_party/docs/architecture/icepanel/models/dlh-in-a-box.json](#) in the DLH-in-a-box chart repository.

9.66. Trino Access-Control Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Generates Trino access-control settings, including file-based rules or Ranger-backed authorization.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/templates/configmap-access-control-coordinator.yaml](#)
- [GitHub: charts/dlh-in-a-box/charts/trino/templates/configmap-access-control-worker.yaml](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Generates Trino access-control settings, including file-based rules or Ranger-backed authorization.

Model metadata: source ID DLH-C3-TRINO-ACCESS; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/trino/templates/configmap-access-control-coordinator.yaml](#); GitHub: [charts/dlh-in-a-box/charts/trino/templates/configmap-access-control-worker.yaml](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.67. Trino Catalog Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Generates Trino catalog settings, mapping chart settings to catalog properties for object-backed datasets.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/templates/configmap-catalog.yaml](https://github.com/dlh-in-a-box/charts/trino/templates/configmap-catalog.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Generates Trino catalog settings, mapping chart settings to catalog properties for object-backed datasets.

Model metadata: source ID DLH-C3-TRINO-CATALOGS; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/trino/templates/configmap-catalog.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.68. Trino Coordinator Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Renders the Trino coordinator deployment and wires identity, credentials, catalog, access-control, and object-storage settings.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/templates/deployment-coordinator.yaml](https://github.com/dlh-in-a-box/charts/trino/templates/deployment-coordinator.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Renders the Trino coordinator deployment and wires identity, credentials, catalog, access-control, and object-storage settings.

Model metadata: source ID DLH-C3-TRINO-COORDINATOR; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/trino/templates/deployment-coordinator.yaml](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.69. Trino Integration Documentation

Internal component in the dlh-in-a-box domain.

Status: live

Explains why the local Trino chart is included and documents the supported integration boundary.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/OVERVIEW.md](#)
- [GitHub: charts/dlh-in-a-box/charts/trino/templates/_README.txt](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Explains why the local Trino chart is included and documents the supported integration boundary.

Model metadata: source ID DLH-C3-TRINO-DOCS; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: [charts/dlh-in-a-box/charts/trino/OVERVIEW.md](#); GitHub: [charts/dlh-in-a-box/charts/trino/templates/_README.txt](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.70. Trino Integration Helpers

Internal component in the dlh-in-a-box domain.

Status: live

Defines shared templating functions used by the local Trino integration.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/templates/_helpers.tpl](https://github.com/wellcome-trino/charts/dlh-in-a-box/charts/trino/templates/_helpers.tpl)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Defines shared templating functions used by the local Trino integration.

Model metadata: source ID DLH-C3-TRINO-HELPERS; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/trino/templates/_helpers.tpl. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.71. Trino Worker Rendering

Internal component in the dlh-in-a-box domain.

Status: live

Renders Trino worker deployments with the identity, credential, catalog, and storage integration required for distributed query execution.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/trino/templates/deployment-worker.yaml](https://github.com/wellcome-trino/charts/dlh-in-a-box/charts/trino/templates/deployment-worker.yaml)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Vendored Trino Chart](#)

Renders Trino worker deployments with the identity, credential, catalog, and storage integration required for distributed query execution.

Model metadata: source ID DLH-C3-TRINO-WORKER; IcePanel type component within a parent object; Child of Vendored Trino Chart (DLH-C2-TRINO-VENDORED). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/trino/templates/deployment-worker.yaml. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.72. Umbrella Templates

Internal component in the dlh-in-a-box domain.

Status: live

Repository-owned Helm templates that adapt and coordinate dependency charts into a coherent runtime.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/templates](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Repository-owned Helm templates that adapt and coordinate dependency charts into a coherent runtime.

Model metadata: source ID DLH-C3-SOURCE-TEMPLATES; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/templates. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.73. Values Contract

Internal component in the dlh-in-a-box domain.

Status: live

Defines the default settings and supported user-facing settings schema used to render the chart.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/values.yaml](#)
- [GitHub: charts/dlh-in-a-box/values.schema.json](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Chart Source Components](#)

Defines the default settings and supported user-facing settings schema used to render the chart.

Model metadata: source ID DLH-C3-SOURCE-VALUES; IcePanel type component within a parent object; Child of Chart Source (DLH-C2-CHART-SOURCE). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: [GitHub: charts/dlh-in-a-box/values.yaml](#); [GitHub: charts/dlh-in-a-box/values.schema.json](#). Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.74. Vault Bootstrap Script

Internal component in the The PaM Data Mesh domain.

Status: live

Initializes or unseals Vault, enables auth methods, applies policies, and writes bootstrap paths.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](#)

Teams:

- Sanger PaM Data Engineering and Integration

- icddr,b IT

In diagrams:

- [Vault Component Diagram](#)

Initializes or unseals Vault, enables auth methods, applies policies, and writes bootstrap paths. Model metadata: source ID CMP-VAULT-BOOTSTRAP; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.75. Vault Chart Archive

Internal component in the dlh-in-a-box domain.

Status: live

Bundled Helm archive for optional Vault secret-management deployment included with the chart.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitHub: charts/dlh-in-a-box/charts/vault-0.32.0.tgz](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Packaged Upstream Dependencies](#)

Bundled Helm archive for optional Vault secret-management deployment included with the chart.

Model metadata: source ID DLH-C3-PKG-VAULT; IcePanel type component within a parent object; Child of Packaged Upstream Dependency Helm Charts (DLH-C2-UPSTREAM-ARCHIVES). No IcePanel group membership is assigned. Lifecycle status: Live in the current IcePanel model. Ownership or stewardship: Sanger PaM Data Engineering and Integration. Location or hosting context: Wellcome Sanger Institute, Cloud. Reality links: GitHub: charts/dlh-in-a-box/charts/vault-0.32.0.tgz. Source of truth: docs/architecture/icepanel/models/dlh-in-a-box.json in the DLH-in-a-box chart repository.

9.76. Vault OIDC Sync

Internal component in the The PaM Data Mesh domain.

Status: live

Configures Vault OIDC/JWT auth against Keycloak or Platform Home.

Locations: Wellcome Sanger Institute, icddr,b

Links:

- [GitHub: platform/dlh-in-a-box/scripts](https://github.com/platform/dlh-in-a-box/scripts)

Teams:

- Sanger PaM Data Engineering and Integration
- icddr,b IT

In diagrams:

- [Vault Component Diagram](#)

Configures Vault OIDC/JWT auth against Keycloak or Platform Home. Model metadata: source ID CMP-VAULT-OIDC; IcePanel type component within a parent object; lifecycle status Live in the current IcePanel model (inferred from parent lifecycle live); ownership or stewardship Sanger PaM Data Engineering and Integration, icddr,b IT; location or hosting context Wellcome Sanger Institute, icddr,b; reality link GitHub: platform/dlh-in-a-box/scripts. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.

9.77. warehouse

Internal component in the The PaM Data Mesh domain.

Status: future

Schema, table, catalog registration, and query helpers.

Locations: Wellcome Sanger Institute, Cloud

Links:

- [GitLab: dlh-database-client package](#)

Teams:

- Sanger PaM Data Engineering and Integration

In diagrams:

- [Shared Framework Package Component Diagram](#)

Schema, table, catalog registration, and query helpers. Model metadata: source ID CMP-FW-WAREHOUSE; IcePanel type component within a parent object; lifecycle status Planned or target-state object (inferred from parent lifecycle future); ownership or stewardship Sanger PaM Data Engineering and Integration; location or hosting context Wellcome Sanger Institute, Cloud; reality link GitLab: dlh-database-client package. Source of truth: docs/architecture/c4-icepanel-model.xml in the icddrb-data-platform-infra repository.